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(54) **SUMMARIZATION AND PERSONALIZATION OF BIG DATA METHOD AND APPARATUS**

(71) Applicant: **Avalara, Inc.**, Seattle, WA (US)

(72) Inventors: **Praveen SELVAM**, Chennai (IN); **Sanjay PARTHASARATHY**, Bellevue, WA (US); **Satyanarayana Rao KALIKIVAYI**, Chennai (IN)

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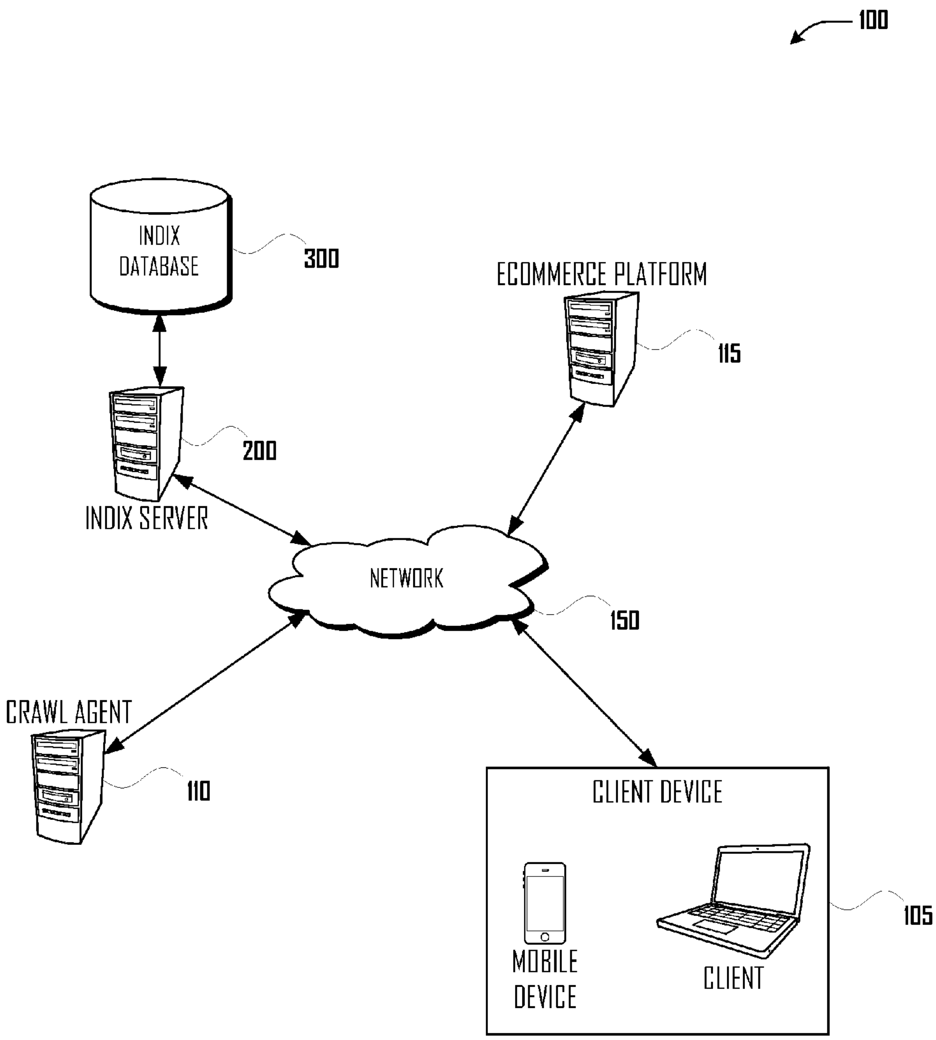
(52) **U.S. Cl.**
CPC ***G06Q 30/0623*** (2013.01); ***G06Q 10/087*** (2013.01); ***G06Q 30/0201*** (2013.01); ***G06Q 30/0202*** (2013.01)

(57) **ABSTRACT**

Systems and methods for a user interface to summarize and personalize a large amount of price and product information, to identify patterns therein, and to generate recommendations in relation thereto are described herein.



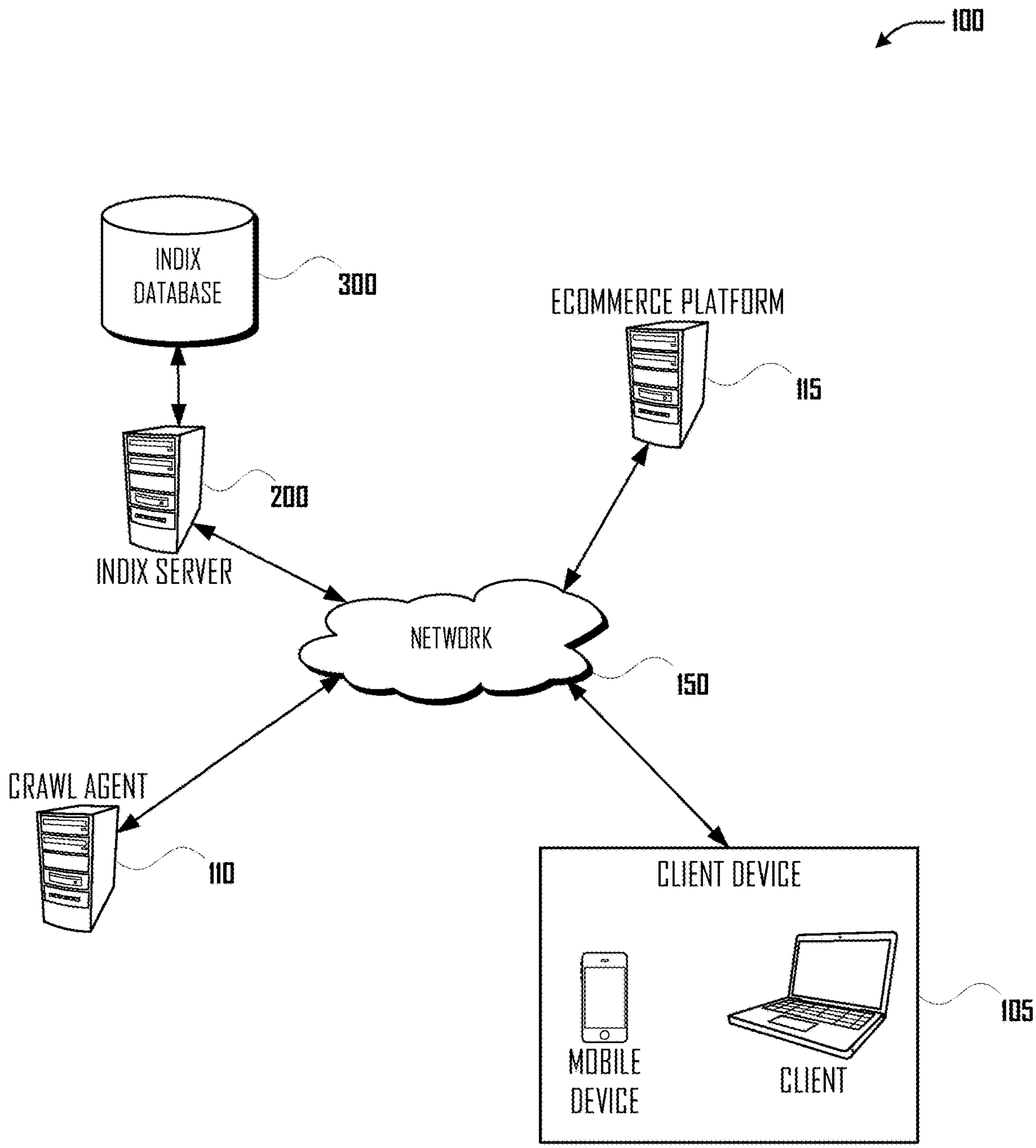

100
INDIX DATABASE
300
ECOMMERCE PLATFORM
115
200
INDIX SERVER
NETWORK
150
CRAWL AGENT
110
CLIENT DEVICE
105
MOBILE DEVICE
CLIENT

Fig.1

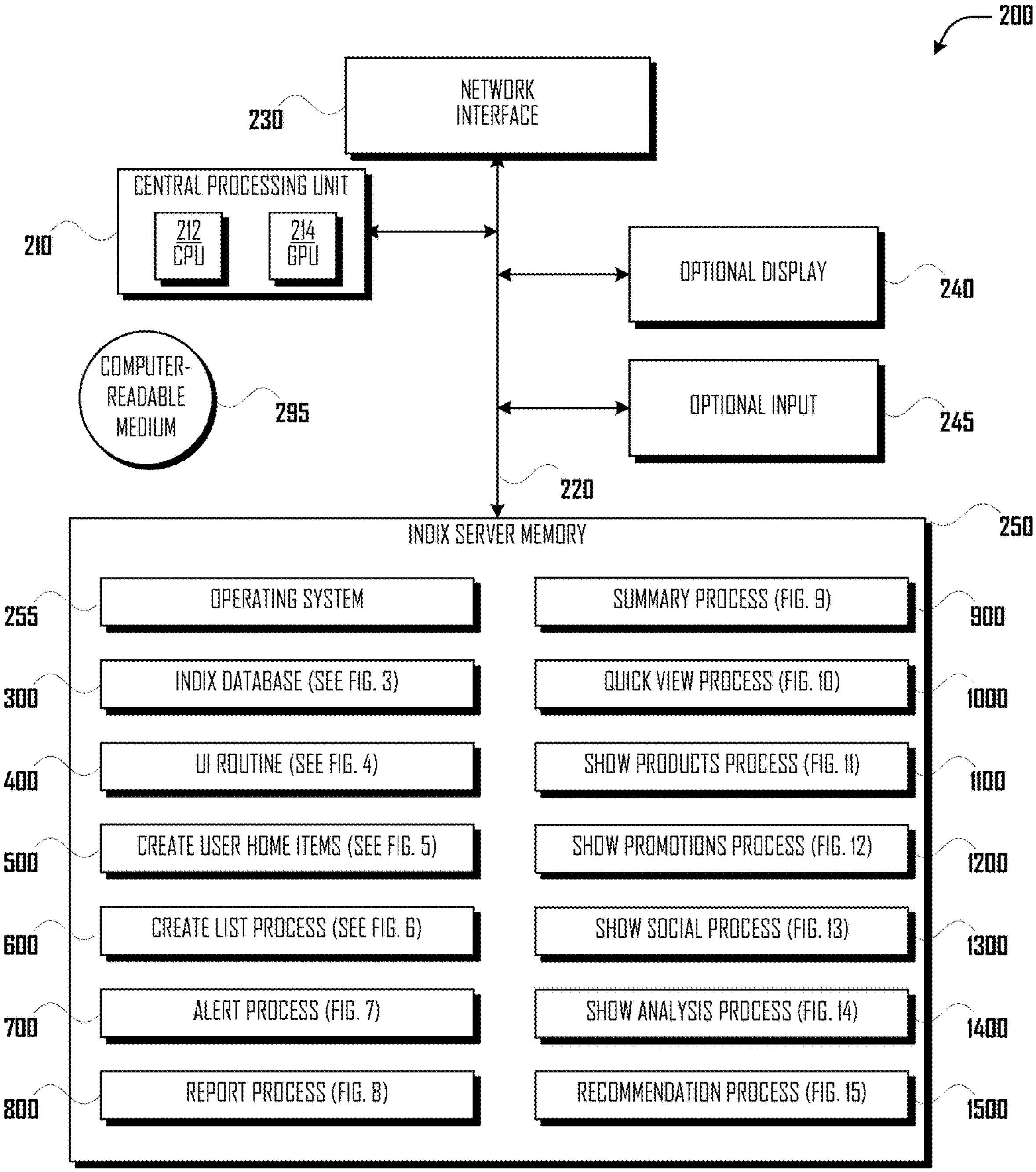

200
NETWORK INTERFACE
230
CENTRAL PROCESSING UNIT
212 CPU
214 GPU
210
OPTIONAL DISPLAY
240
OPTIONAL INPUT
245
COMPUTER-READABLE MEDIUM
295
220
INDIX SERVER MEMORY
250
OPERATING SYSTEM
255
INDIX DATABASE (SEE FIG. 3)
300
UI ROUTINE (SEE FIG. 4)
400
CREATE USER HOME ITEMS (SEE FIG. 5)
500
CREATE LIST PROCESS (SEE FIG. 6)
600
ALERT PROCESS (FIG. 7)
700
REPORT PROCESS (FIG. 8)
800
SUMMARY PROCESS (FIG. 9)
900
QUICK VIEW PROCESS (FIG. 10)
1000
SHOW PRODUCTS PROCESS (FIG. 11)
1100
SHOW PROMOTIONS PROCESS (FIG. 12)
1200
SHOW SOCIAL PROCESS (FIG. 13)
1300
SHOW ANALYSIS PROCESS (FIG. 14)
1400
RECOMMENDATION PROCESS (FIG. 15)
1500

Fig. 2

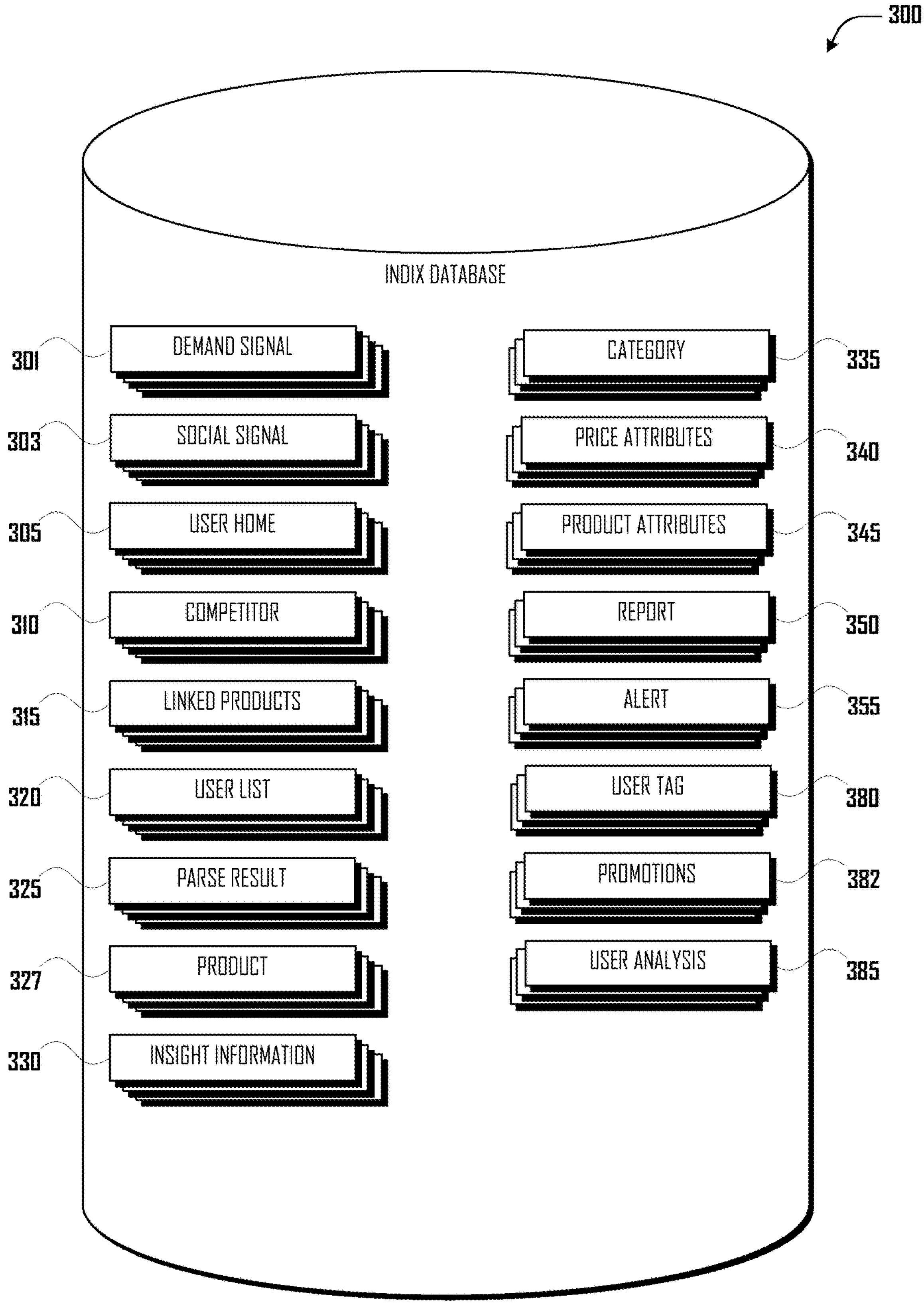

300
INDIX DATABASE
301
DEMAND SIGNAL
303
SOCIAL SIGNAL
305
USER HOME
310
COMPETITOR
315
LINKED PRODUCTS
320
USER LIST
325
PARSE RESULT
327
PRODUCT
330
INSIGHT INFORMATION
CATEGORY
335
PRICE ATTRIBUTES
340
PRODUCT ATTRIBUTES
345
REPORT
350
ALERT
355
USER TAG
380
PROMOTIONS
382
USER ANALYSIS
385

Fig.3

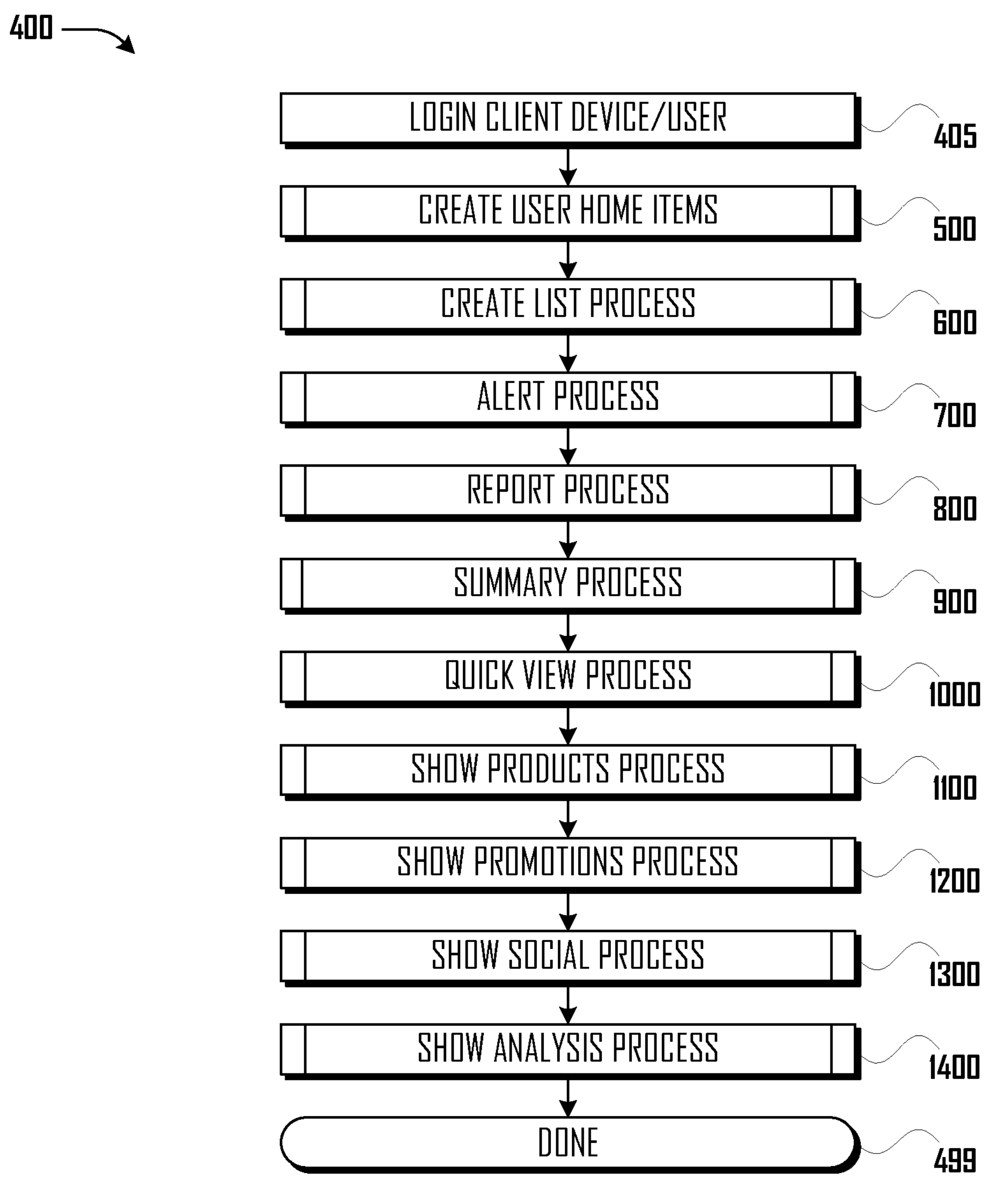

400
LOGIN CLIENT DEVICE/USER
405
CREATE USER HOME ITEMS
500
CREATE LIST PROCESS
600
ALERT PROCESS
700
REPORT PROCESS
800
SUMMARY PROCESS
900
QUICK VIEW PROCESS
1000
SHOW PRODUCTS PROCESS
1100
SHOW PROMOTIONS PROCESS
1200
SHOW SOCIAL PROCESS
1300
SHOW ANALYSIS PROCESS
1400
DONE
499

Fig. 4

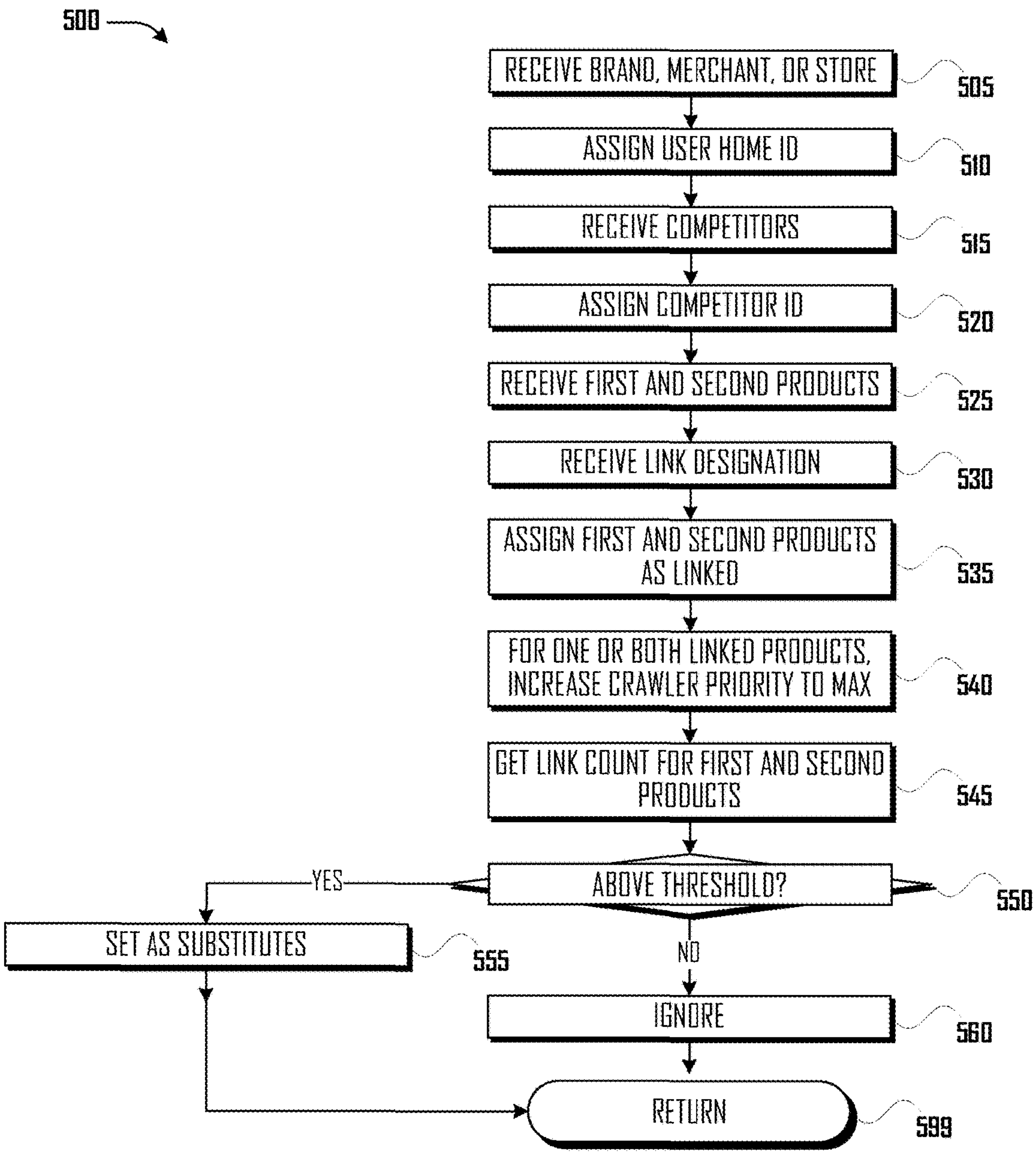

500
RECEIVE BRAND, MERCHANT, OR STORE
505
ASSIGN USER HOME ID
510
RECEIVE COMPETITORS
515
ASSIGN COMPETITOR ID
520
RECEIVE FIRST AND SECOND PRODUCTS
525
RECEIVE LINK DESIGNATION
530
ASSIGN FIRST AND SECOND PRODUCTS AS LINKED
535
FOR ONE OR BOTH LINKED PRODUCTS, INCREASE CRAWLER PRIORITY TO MAX
540
GET LINK COUNT FOR FIRST AND SECOND PRODUCTS
545
ABOVE THRESHOLD?
550
YES
SET AS SUBSTITUTES
555
NO
IGNORE
560
RETURN
599

Fig. 5

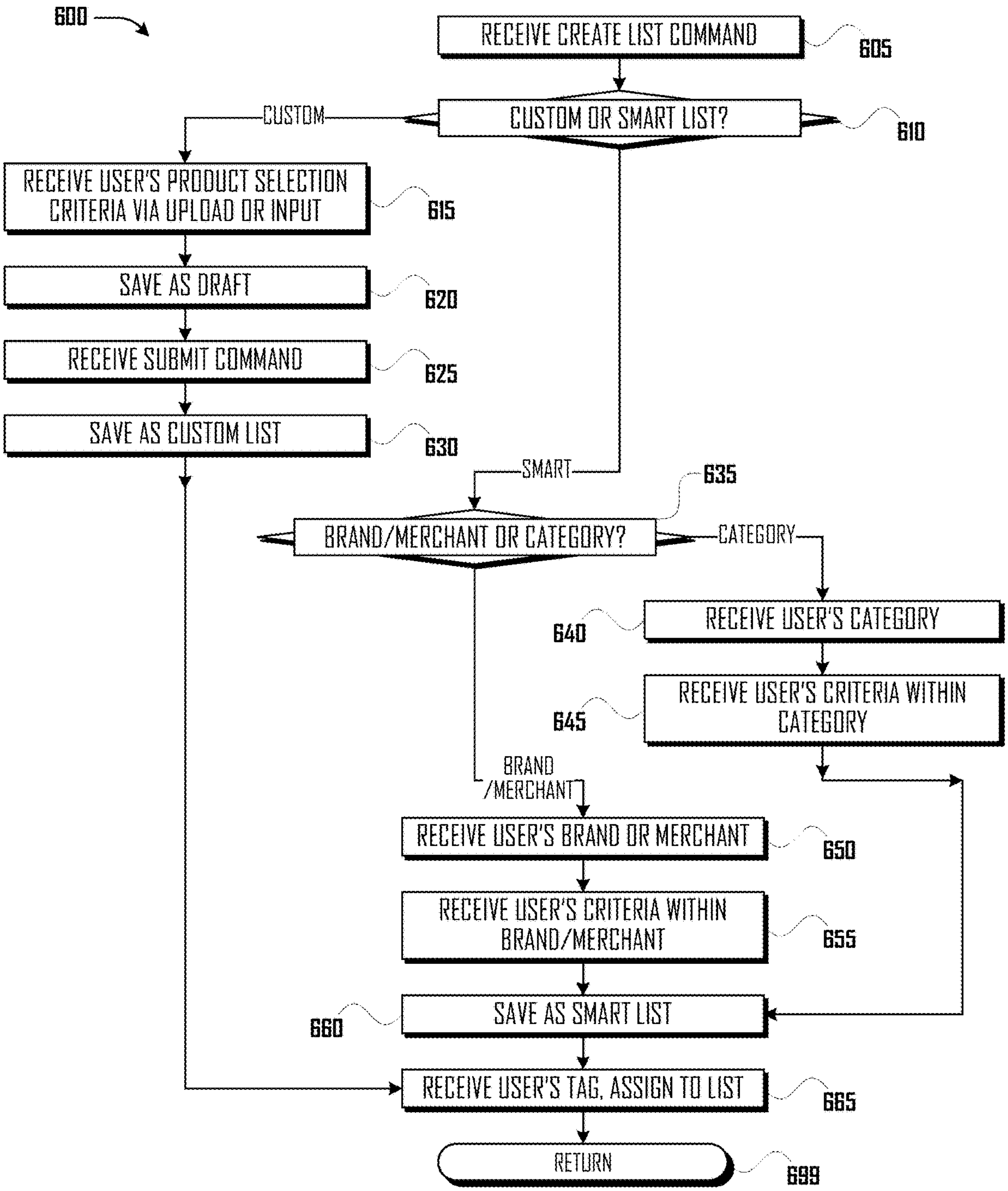

600
RECEIVE CREATE LIST COMMAND
605
CUSTOM OR SMART LIST?
610
CUSTOM
RECEIVE USER'S PRODUCT SELECTION CRITERIA VIA UPLOAD OR INPUT
615
SAVE AS DRAFT
620
RECEIVE SUBMIT COMMAND
625
SAVE AS CUSTOM LIST
630
SMART
BRAND/MERCHANT OR CATEGORY?
635
CATEGORY
RECEIVE USER'S CATEGORY
640
RECEIVE USER'S CRITERIA WITHIN CATEGORY
645
BRAND /MERCHANT
RECEIVE USER'S BRAND OR MERCHANT
650
RECEIVE USER'S CRITERIA WITHIN BRAND/MERCHANT
655
SAVE AS SMART LIST
660
RECEIVE USER'S TAG, ASSIGN TO LIST
665
RETURN
699

Fig. 6

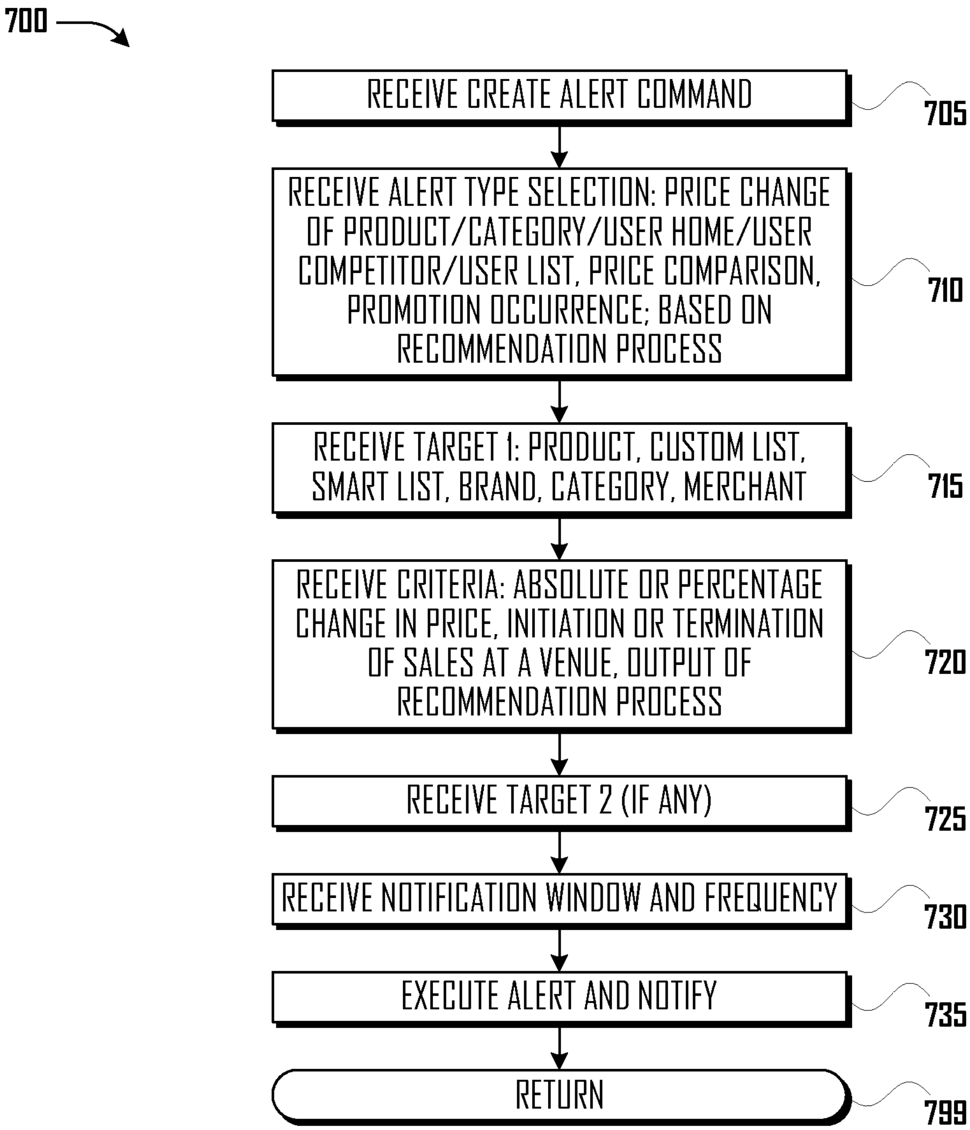

700
RECEIVE CREATE ALERT COMMAND
705
RECEIVE ALERT TYPE SELECTION: PRICE CHANGE OF PRODUCT/CATEGORY/USER HOME/USER COMPETITOR/USER LIST, PRICE COMPARISON, PROMOTION OCCURRENCE; BASED ON RECOMMENDATION PROCESS
710
RECEIVE TARGET 1: PRODUCT, CUSTOM LIST, SMART LIST, BRAND, CATEGORY, MERCHANT
715
RECEIVE CRITERIA: ABSOLUTE OR PERCENTAGE CHANGE IN PRICE, INITIATION OR TERMINATION OF SALES AT A VENUE, OUTPUT OF RECOMMENDATION PROCESS
720
RECEIVE TARGET 2 (IF ANY)
725
RECEIVE NOTIFICATION WINDOW AND FREQUENCY
730
EXECUTE ALERT AND NOTIFY
735
RETURN
799

Fig. 7

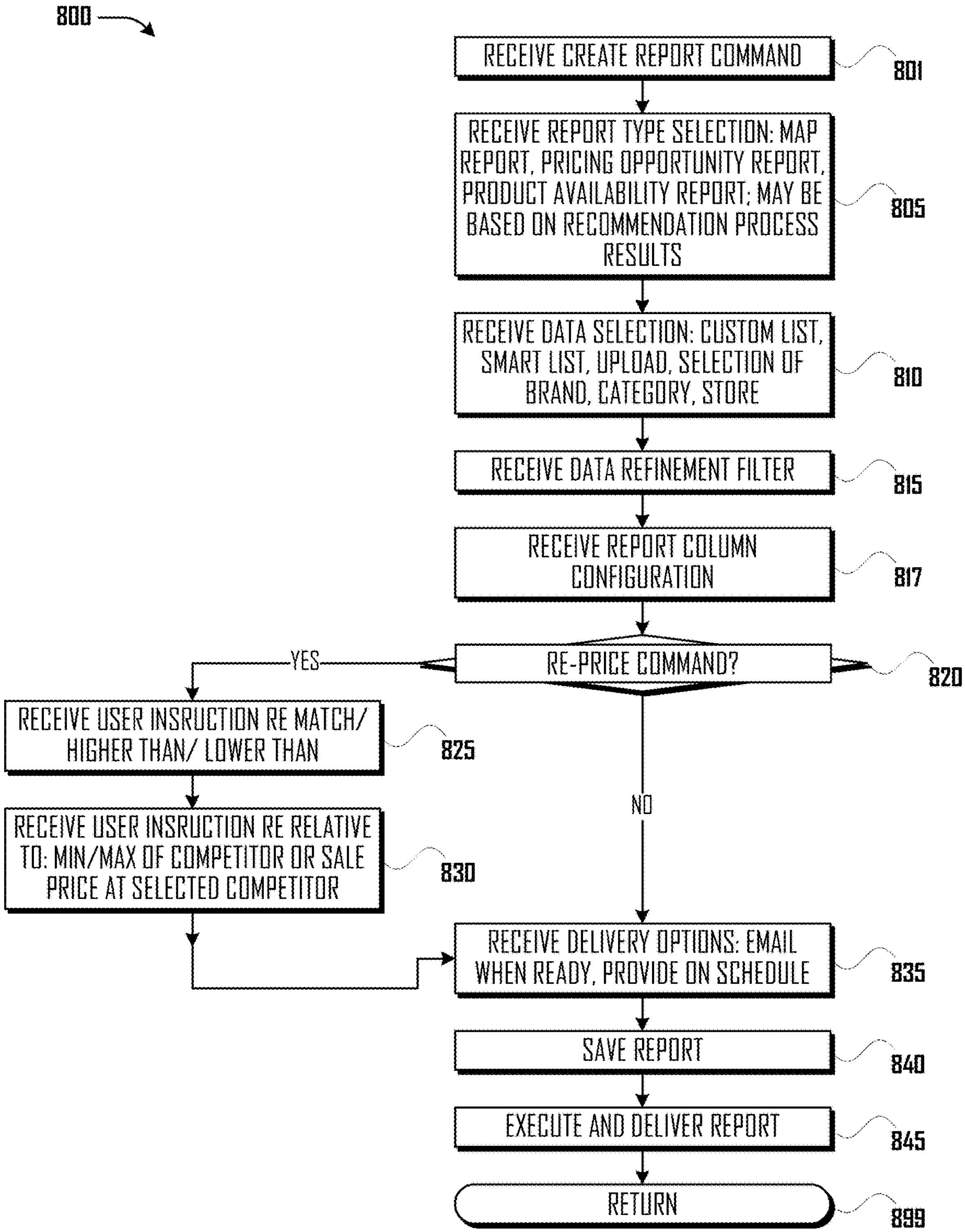

800
RECEIVE CREATE REPORT COMMAND
801
RECEIVE REPORT TYPE SELECTION: MAP REPORT, PRICING OPPORTUNITY REPORT, PRODUCT AVAILABILITY REPORT; MAY BE BASED ON RECOMMENDATION PROCESS RESULTS
805
RECEIVE DATA SELECTION: CUSTOM LIST, SMART LIST, UPLOAD, SELECTION OF BRAND, CATEGORY, STORE
810
RECEIVE DATA REFINEMENT FILTER
815
RECEIVE REPORT COLUMN CONFIGURATION
817
RE-PRICE COMMAND?
820
YES
RECEIVE USER INSRUCTION RE MATCH/ HIGHER THAN/ LOWER THAN
825
RECEIVE USER INSRUCTION RE RELATIVE TO: MIN/MAX OF COMPETITOR OR SALE PRICE AT SELECTED COMPETITOR
830
NO
RECEIVE DELIVERY OPTIONS: EMAIL WHEN READY, PROVIDE ON SCHEDULE
835
SAVE REPORT
840
EXECUTE AND DELIVER REPORT
845
RETURN
899

Fig. 8

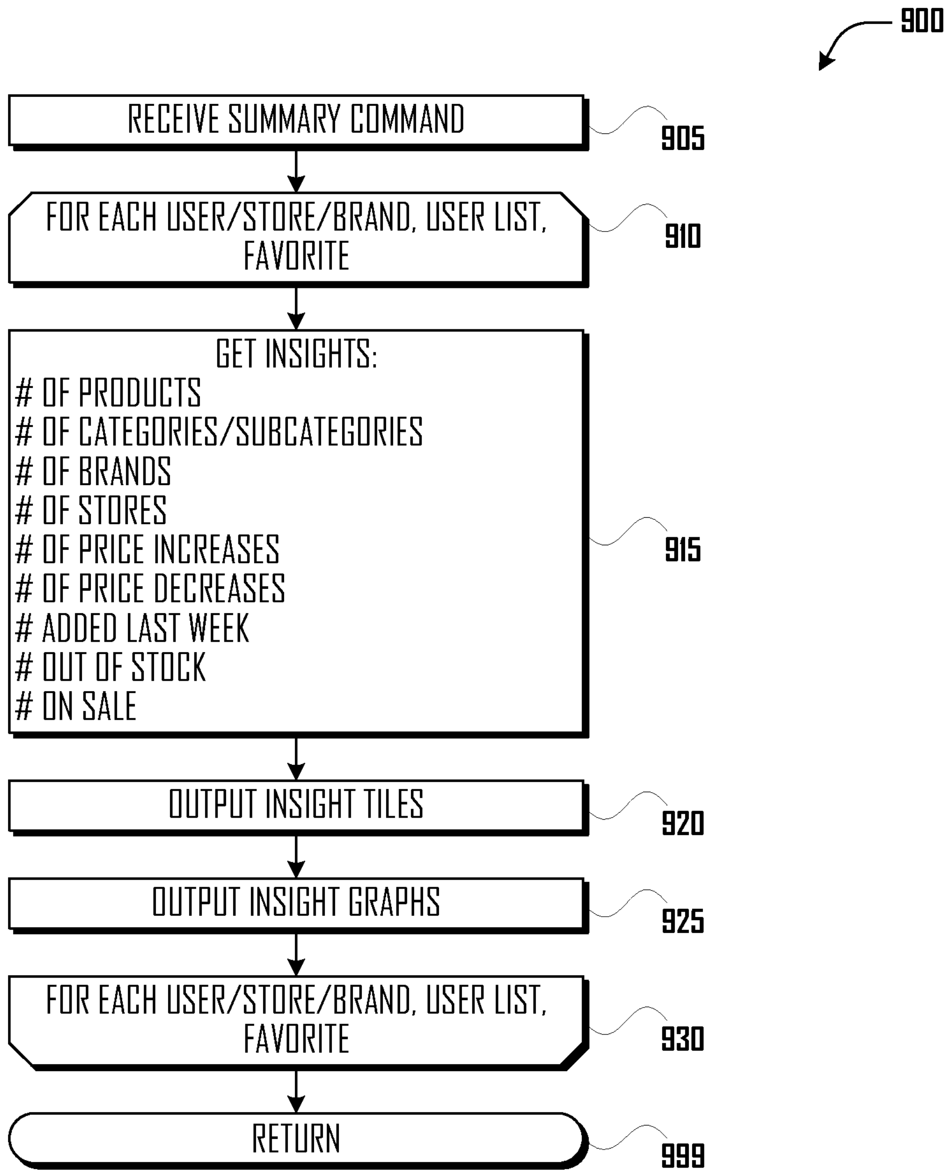

900
RECEIVE SUMMARY COMMAND
905
FOR EACH USER/STORE/BRAND, USER LIST, FAVORITE
910
GET INSIGHTS:
OF PRODUCTS
OF CATEGORIES/SUBCATEGORIES
OF BRANDS
OF STORES
OF PRICE INCREASES
OF PRICE DECREASES
ADDED LAST WEEK
OUT OF STOCK
ON SALE
915
OUTPUT INSIGHT TILES
920
OUTPUT INSIGHT GRAPHS
925
FOR EACH USER/STORE/BRAND, USER LIST, FAVORITE
930
RETURN
999

Fig.9

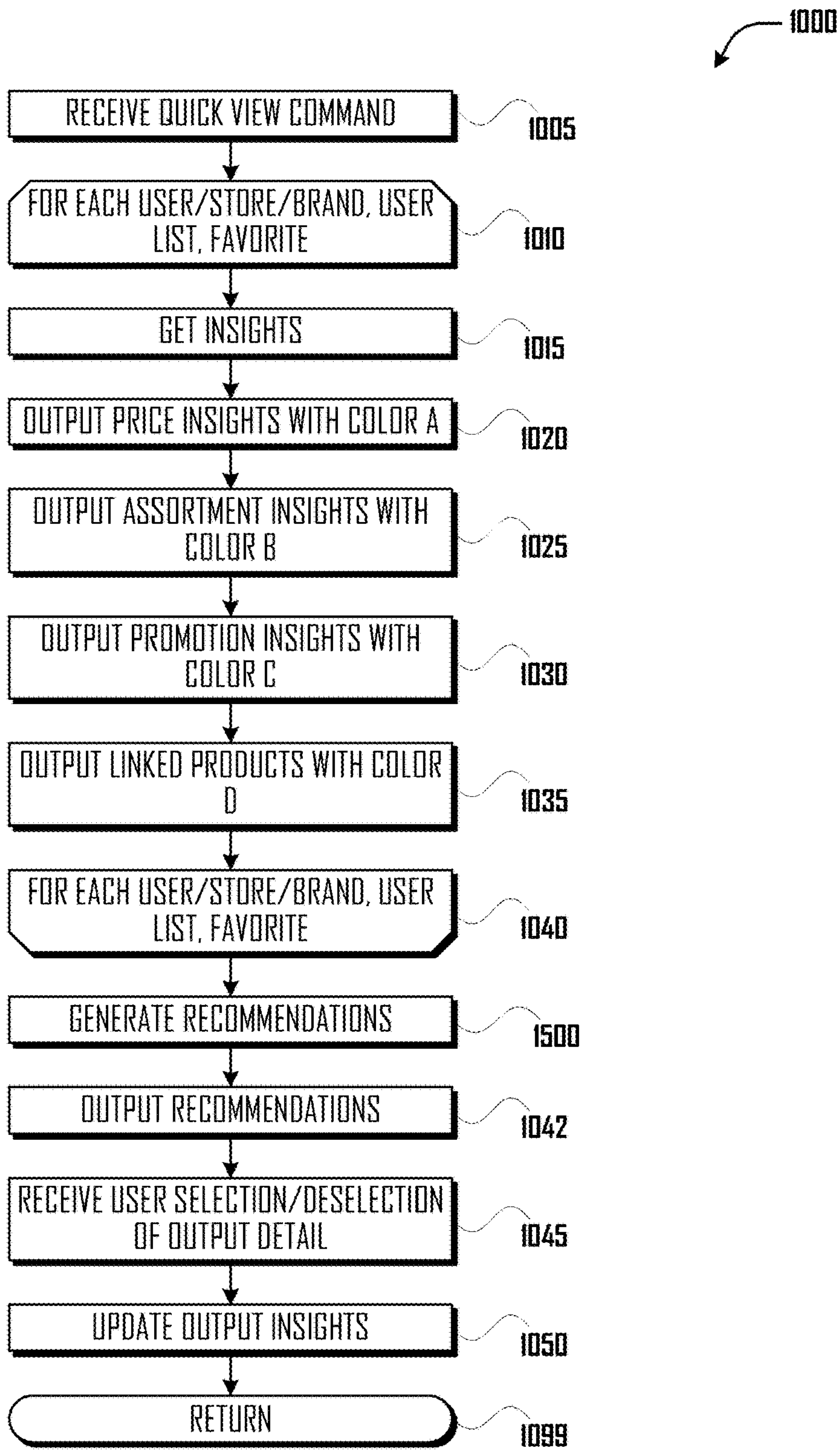

1000
RECEIVE QUICK VIEW COMMAND
1005
FOR EACH USER/STORE/BRAND, USER LIST, FAVORITE
1010
GET INSIGHTS
1015
OUTPUT PRICE INSIGHTS WITH COLOR A
1020
OUTPUT ASSORTMENT INSIGHTS WITH COLOR B
1025
OUTPUT PROMOTION INSIGHTS WITH COLOR C
1030
OUTPUT LINKED PRODUCTS WITH COLOR D
1035
FOR EACH USER/STORE/BRAND, USER LIST, FAVORITE
1040
GENERATE RECOMMENDATIONS
1500
OUTPUT RECOMMENDATIONS
1042
RECEIVE USER SELECTION/DESELECTION OF OUTPUT DETAIL
1045
UPDATE OUTPUT INSIGHTS
1050
RETURN
1099

Fig. 10

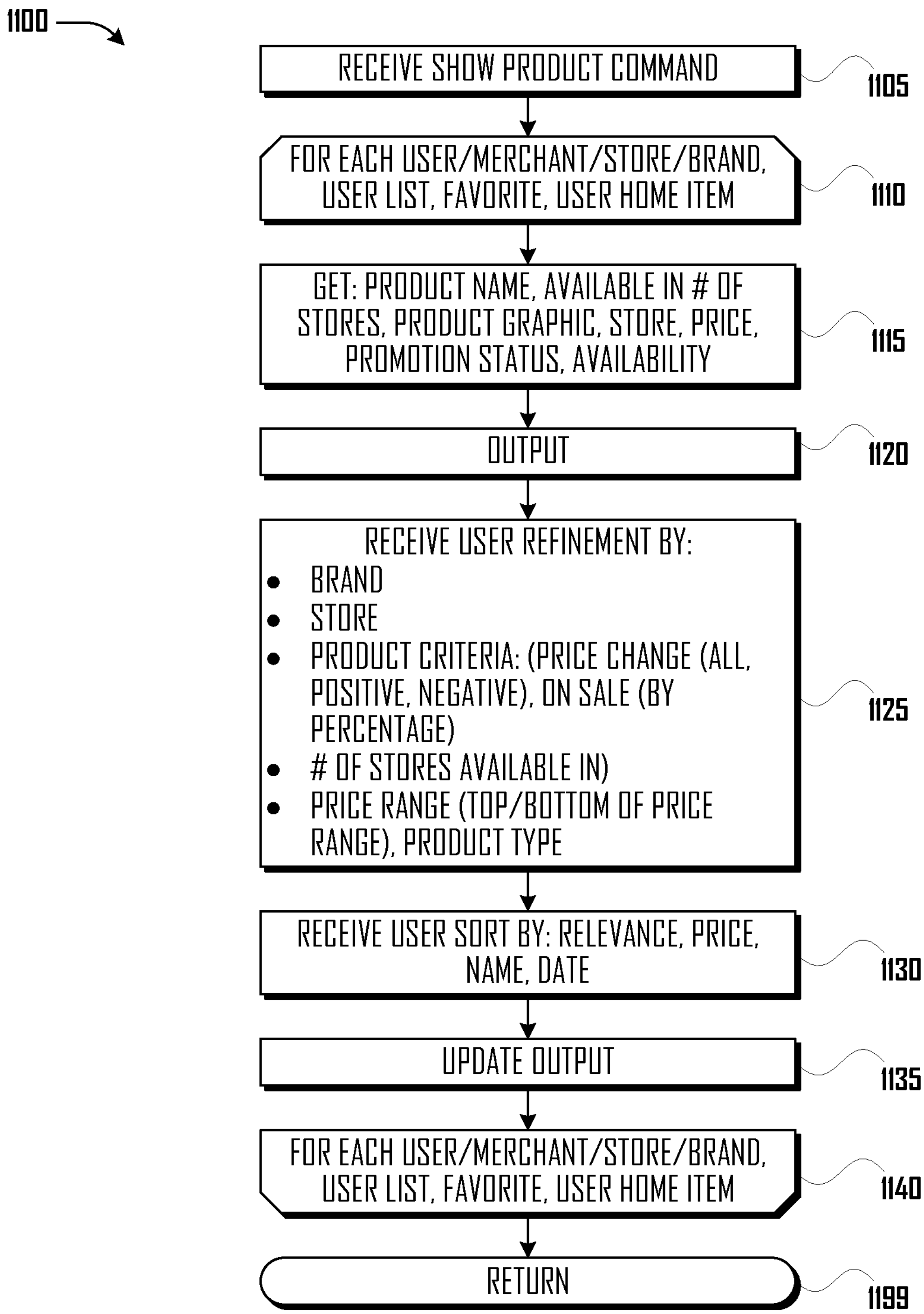

1100
RECEIVE SHOW PRODUCT COMMAND
1105
FOR EACH USER/MERCHANT/STORE/BRAND, USER LIST, FAVORITE, USER HOME ITEM
1110
GET: PRODUCT NAME, AVAILABLE IN # OF STORES, PRODUCT GRAPHIC, STORE, PRICE, PROMOTION STATUS, AVAILABILITY
1115
OUTPUT
1120
RECEIVE USER REFINEMENT BY:
• BRAND
• STORE
• PRODUCT CRITERIA: (PRICE CHANGE (ALL, POSITIVE, NEGATIVE), ON SALE (BY PERCENTAGE)
• # OF STORES AVAILABLE IN)
• PRICE RANGE (TOP/BOTTOM OF PRICE RANGE), PRODUCT TYPE
1125
RECEIVE USER SORT BY: RELEVANCE, PRICE, NAME, DATE
1130
UPDATE OUTPUT
1135
FOR EACH USER/MERCHANT/STORE/BRAND, USER LIST, FAVORITE, USER HOME ITEM
1140
RETURN
1199

Fig. 11

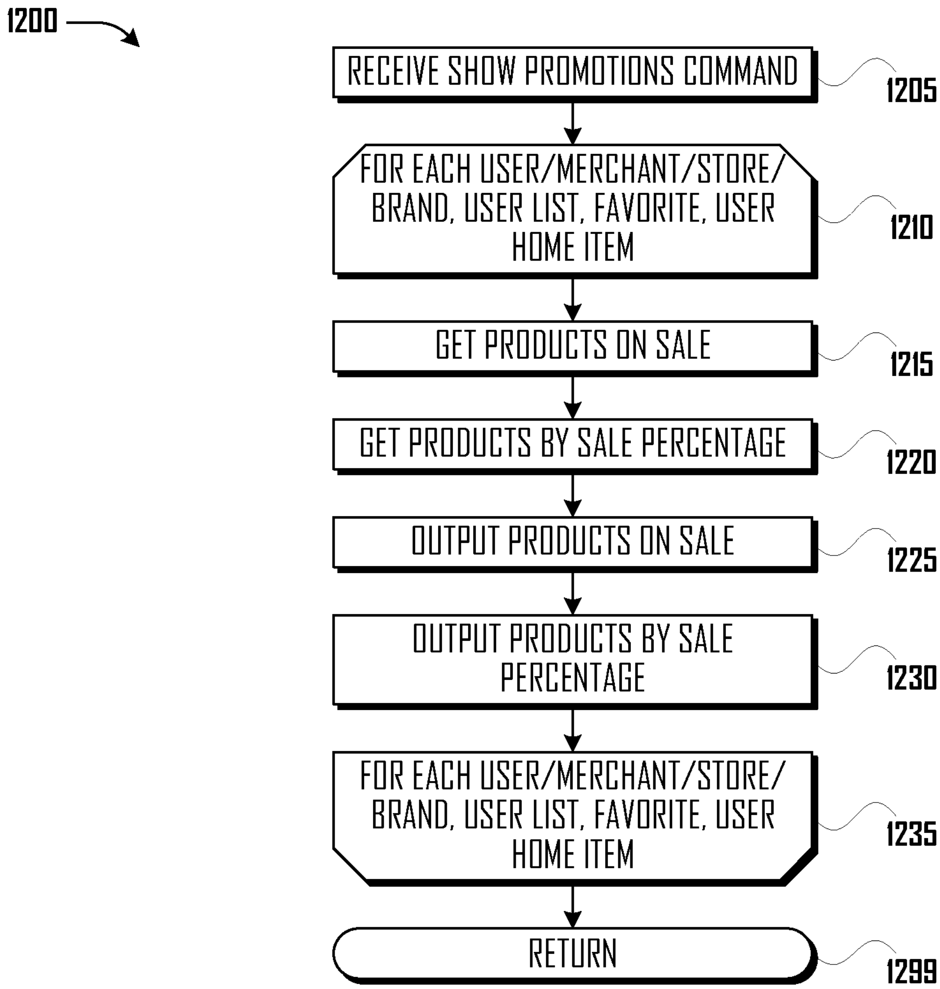

1200
RECEIVE SHOW PROMOTIONS COMMAND
1205
FOR EACH USER/MERCHANT/STORE/BRAND, USER LIST, FAVORITE, USER HOME ITEM
1210
GET PRODUCTS ON SALE
1215
GET PRODUCTS BY SALE PERCENTAGE
1220
OUTPUT PRODUCTS ON SALE
1225
OUTPUT PRODUCTS BY SALE PERCENTAGE
1230
FOR EACH USER/MERCHANT/STORE/BRAND, USER LIST, FAVORITE, USER HOME ITEM
1235
RETURN
1299

Fig.12

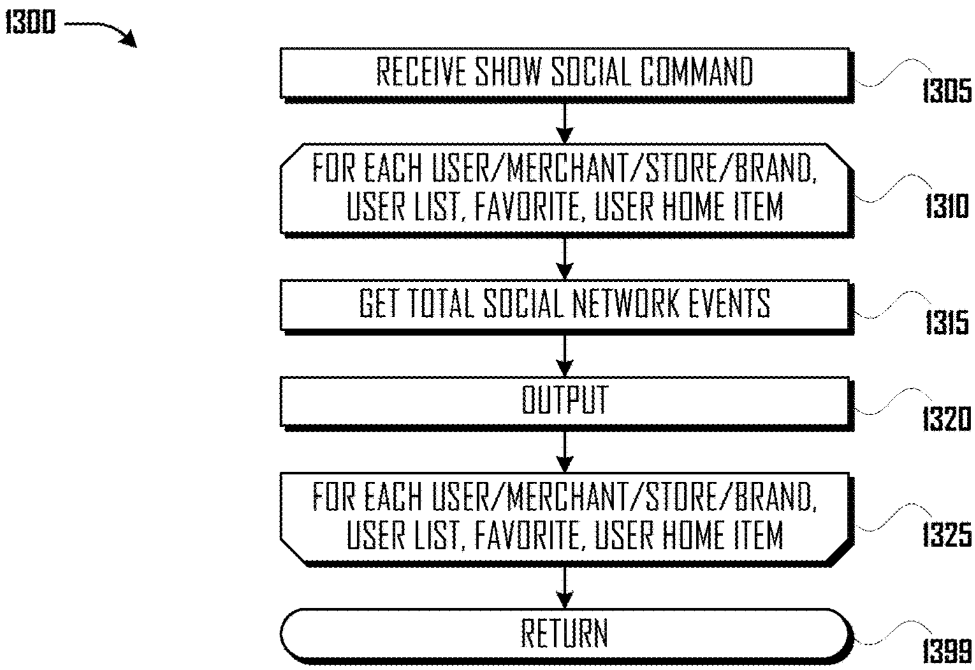

1300
RECEIVE SHOW SOCIAL COMMAND
1305
FOR EACH USER/MERCHANT/STORE/BRAND, USER LIST, FAVORITE, USER HOME ITEM
1310
GET TOTAL SOCIAL NETWORK EVENTS
1315
OUTPUT
1320
FOR EACH USER/MERCHANT/STORE/BRAND, USER LIST, FAVORITE, USER HOME ITEM
1325
RETURN
1399

Fig. 13

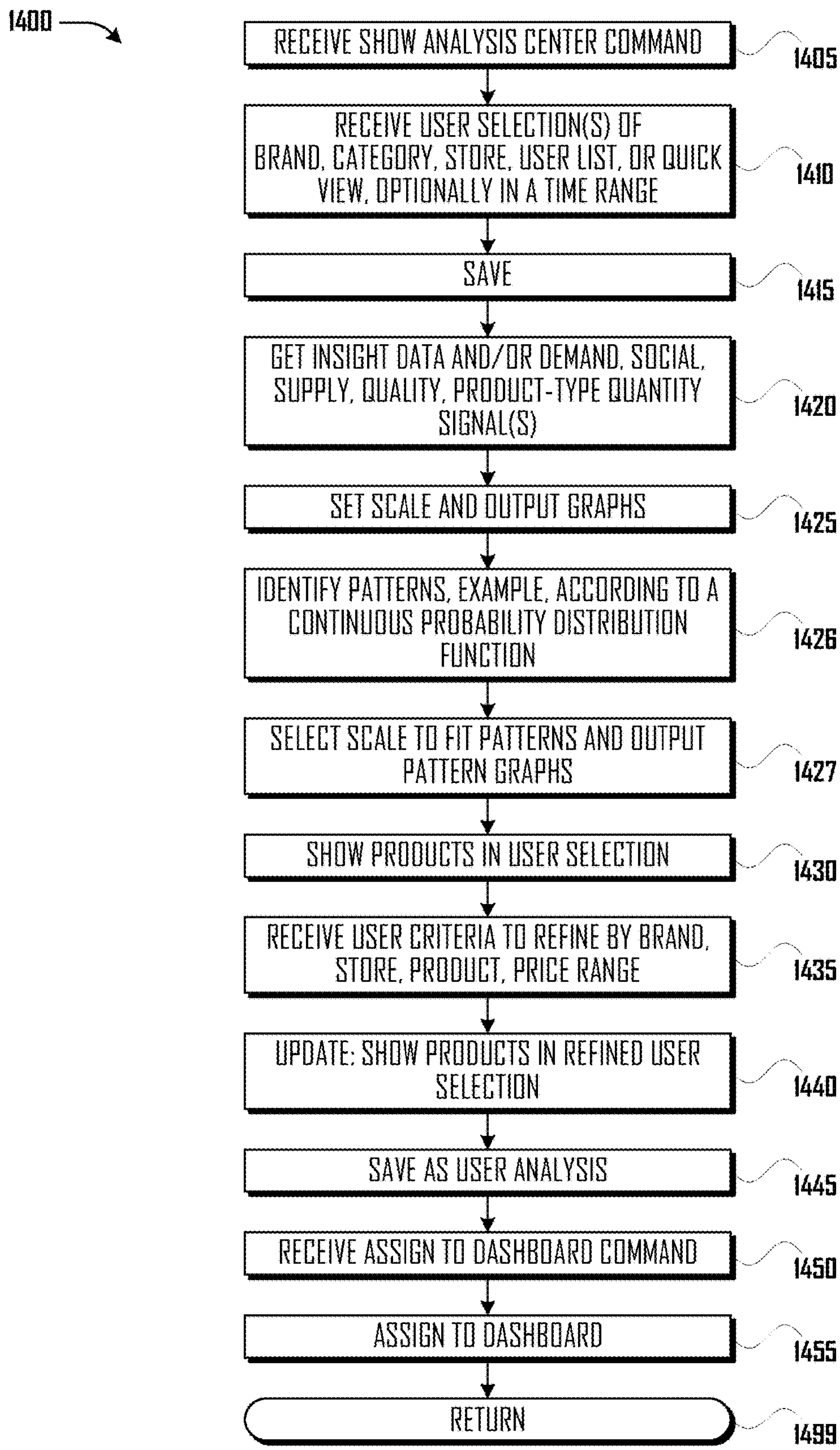

1400
RECEIVE SHOW ANALYSIS CENTER COMMAND
1405
RECEIVE USER SELECTION(S) OF BRAND, CATEGORY, STORE, USER LIST, OR QUICK VIEW, OPTIONALLY IN A TIME RANGE
1410
SAVE
1415
GET INSIGHT DATA AND/OR DEMAND, SOCIAL, SUPPLY, QUALITY, PRODUCT-TYPE QUANTITY SIGNAL(S)
1420
SET SCALE AND OUTPUT GRAPHS
1425
IDENTIFY PATTERNS, EXAMPLE, ACCORDING TO A CONTINUOUS PROBABILITY DISTRIBUTION FUNCTION
1426
SELECT SCALE TO FIT PATTERNS AND OUTPUT PATTERN GRAPHS
1427
SHOW PRODUCTS IN USER SELECTION
1430
RECEIVE USER CRITERIA TO REFINE BY BRAND, STORE, PRODUCT, PRICE RANGE
1435
UPDATE: SHOW PRODUCTS IN REFINED USER SELECTION
1440
SAVE AS USER ANALYSIS
1445
RECEIVE ASSIGN TO DASHBOARD COMMAND
1450
ASSIGN TO DASHBOARD
1455
RETURN
1499

Fig. 14

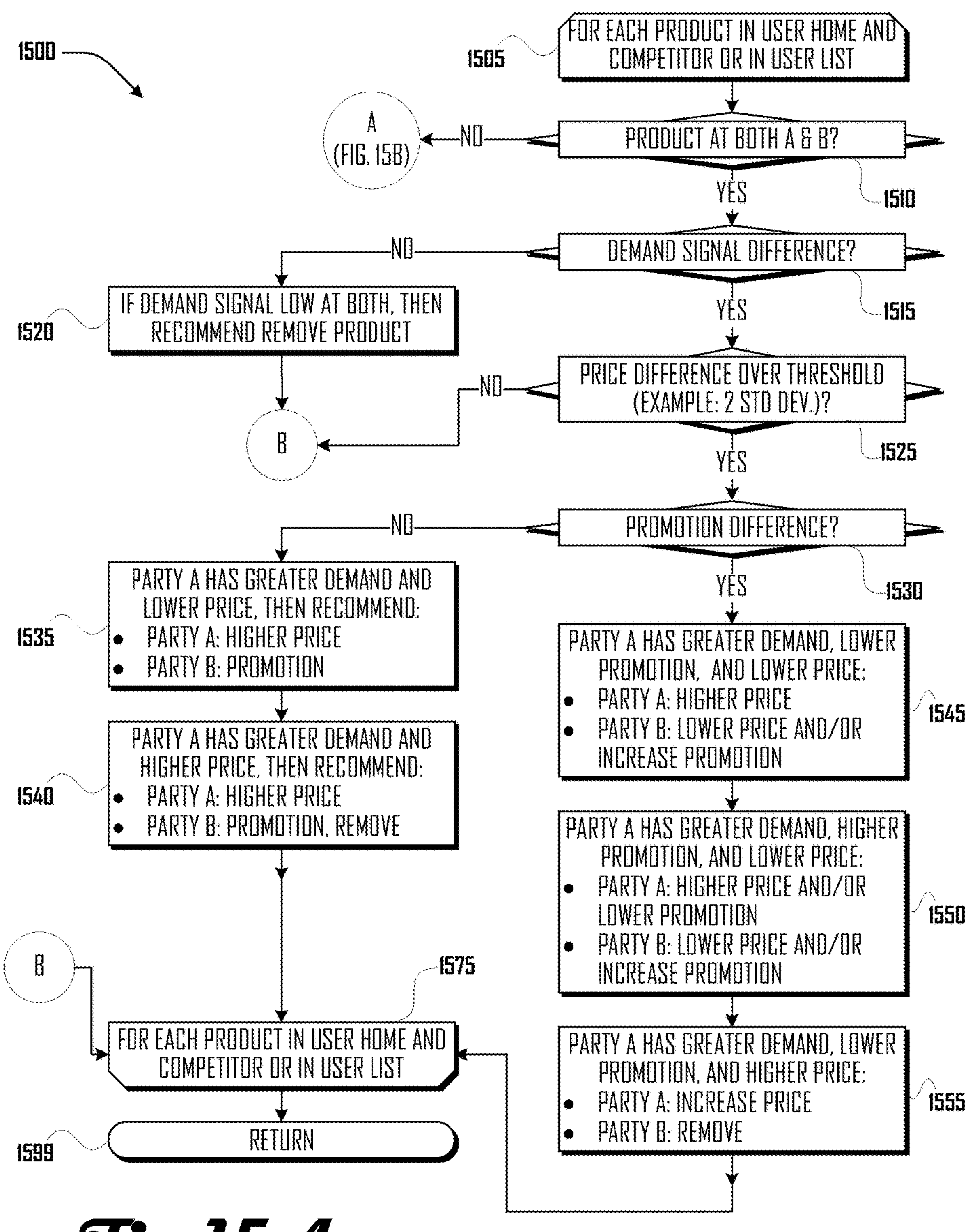

1500
1505
FOR EACH PRODUCT IN USER HOME AND COMPETITOR OR IN USER LIST
PRODUCT AT BOTH A & B?
1510
NO
A (FIG. 15B)
YES
DEMAND SIGNAL DIFFERENCE?
1515
NO
1520
IF DEMAND SIGNAL LOW AT BOTH, THEN RECOMMEND REMOVE PRODUCT
YES
PRICE DIFFERENCE OVER THRESHOLD (EXAMPLE: 2 STD DEV.)?
1525
NO
B
YES
PROMOTION DIFFERENCE?
1530
NO
1535
PARTY A HAS GREATER DEMAND AND LOWER PRICE, THEN RECOMMEND:
• PARTY A: HIGHER PRICE
• PARTY B: PROMOTION
1540
PARTY A HAS GREATER DEMAND AND HIGHER PRICE, THEN RECOMMEND:
• PARTY A: HIGHER PRICE
• PARTY B: PROMOTION, REMOVE
YES
1545
PARTY A HAS GREATER DEMAND, LOWER PROMOTION, AND LOWER PRICE:
• PARTY A: HIGHER PRICE
• PARTY B: LOWER PRICE AND/OR INCREASE PROMOTION
1550
PARTY A HAS GREATER DEMAND, HIGHER PROMOTION, AND LOWER PRICE:
• PARTY A: HIGHER PRICE AND/OR LOWER PROMOTION
• PARTY B: LOWER PRICE AND/OR INCREASE PROMOTION
1555
PARTY A HAS GREATER DEMAND, LOWER PROMOTION, AND HIGHER PRICE:
• PARTY A: INCREASE PRICE
• PARTY B: REMOVE
B
1575
FOR EACH PRODUCT IN USER HOME AND COMPETITOR OR IN USER LIST
1599
RETURN

Fig. 15A

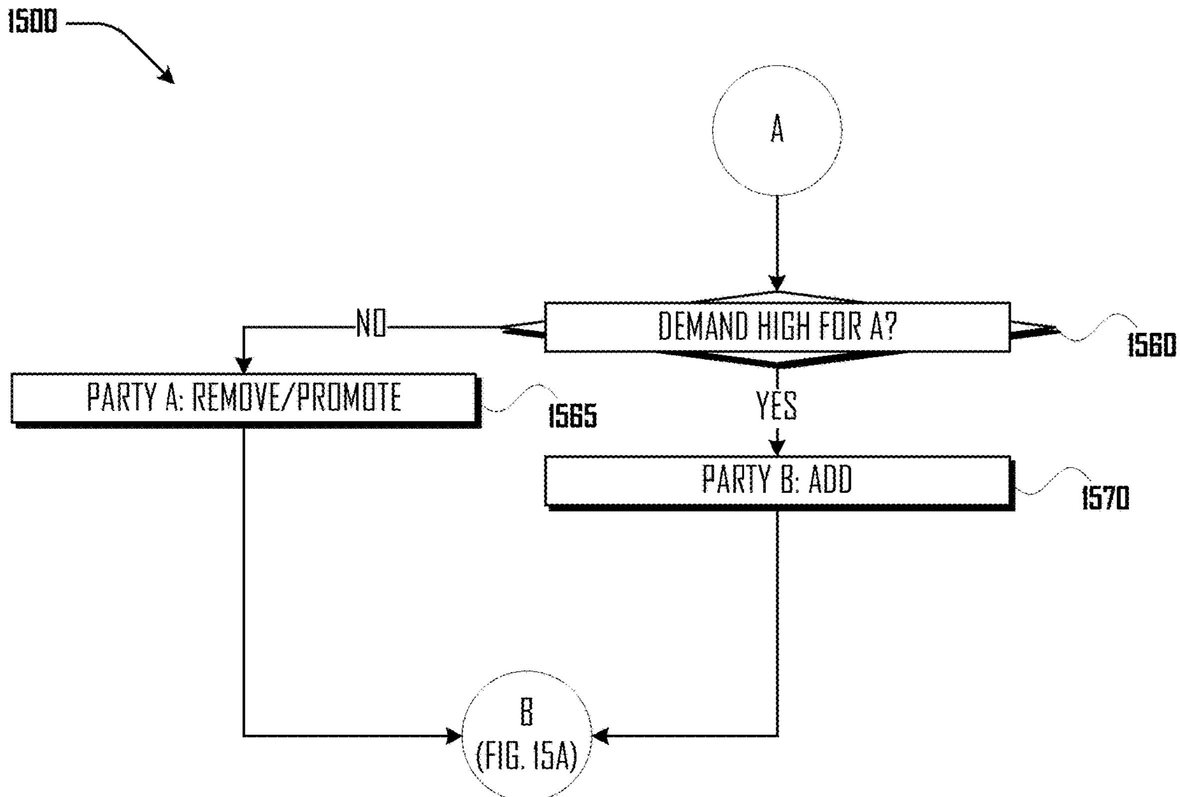

1500
A
DEMAND HIGH FOR A?
1560
NO
YES
PARTY A: REMOVE/PROMOTE
1565
PARTY B: ADD
1570
B
(FIG. 15A)

Fig. 15B

SUMMARIZATION AND PERSONALIZATION OF BIG DATA METHOD AND APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of US Provisional Patent Application Ser. No. 61/952,004, filed on Mar. 12, 2014, and US Provisional Patent Application Ser. No. 61/952,029, filed on Mar. 12, 2014, and is a continuation-in-part of U.S. application Ser. No. 13/951,248, filed Jul. 25, 2013, PCT Application Number PCT/US13/52108, filed Jul. 25, 2013, U.S. application Ser. No. 13/951,244, filed Jul. 25, 2013, and PCT Application Number PCT/US13/52106, filed Jul. 25, 2013, which applications claim the benefit of U.S. Provisional Application Ser. No. 61/675,492, filed on Jul. 25, 2012. All of the foregoing applications and child applications thereof are incorporated herein, in their entirety, for all purposes.

FIELD

[0002] This disclosure is directed to the field of software, and more particularly, to a user interface for a computer, which user interface summarizes and personalizes a very large set of data relating to price and product information.

BACKGROUND

[0003] Indix Corporation collects and analyses a large amount of longitudinal data regarding products, product attributes, prices, and price attributes. To be understood by a person, this large amount of data and analytic output derived therefrom must be summarized, personalized, and organized in relevant terms. The summarization and personalization of such a large and complex set of data presents challenges in the selection and refinement of information as well as with respect to identification of patterns and arrangement of information in a user interface.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** illustrates a computing environment configured according to an embodiment.

[0005] FIG. **2** illustrates an embodiment of Indix Server.

[0006] FIG. **3** illustrates an embodiment of Indix Server Database.

[0007] FIG. **4** illustrates an embodiment of User Interface Process.

[0008] FIG. **5** illustrates an embodiment of Create User Home Items.

[0009] FIG. **6** illustrates an embodiment of Create List Process.

[0010] FIG. **7** illustrates an embodiment of Alert Process.

[0011] FIG. **8** illustrates an embodiment of Report Process.

[0012] FIG. **9** illustrates an embodiment of Summary Process.

[0013] FIG. **10** illustrates an embodiment of Quick View Process.

[0014] FIG. **11** illustrates an embodiment of Show Products Process.

[0015] FIG. **12** illustrates an embodiment of Show Promotions Process.

[0016] FIG. **13** illustrates an embodiment of Show Social Process.

[0017] FIG. **14** illustrates an embodiment of Show Analysis Process.

[0018] FIG. **15**A illustrates a portion of an embodiment of Recommendation Process.

[0019] FIG. **15**B illustrates a portion of an embodiment of Recommendation Process.

DESCRIPTION

[0020] As described further herein, UI Routine **400** allows users to pick a Brand or Merchant (referred to as "User Home **305**"), one or more competitors thereof (referred to as "Competitor **310**"), and to create lists of products and categories of products (referred to as "User Lists **320**"). UI Routine **400** selectively displays information regarding and comparisons among User Home **305**, Competitor **310**, User Lists and other user selections in relation to Products **327**, Categories **335** of Products **327**, Brands, and Merchants. The information regarding these items may be obtained from "Insights." Briefly (a longer description is provided herein), Insights are analytic processes which identify, for example, price changes, and what product and price attributes across the datasets are associated with changes in price. Selections and arrangements of information may be saved to a personalized dashboard.

[0021] Throughout UI Routine **400**, colors are used in relation to various information items, to aid the user in quickly ascertaining, by color, what information is being displayed. For example (in other embodiments, different colors or other approaches to distinguishing items may be used), Brands are orange, Merchants are green, categories are yellow, quickviews are red, smart lists are violet, custom lists are blue, and alerts are black.

[0022] In various embodiments, UI Routine **400** performs one or more of the following processes: Create User Home Items **500**, Create List Process **600**, Alert Process **700**, Report Process **800**, Summary Process **900**, Quick View Process **1000**, Show Products Process **1100**, Show Promotions Process **1200**, Show Social Process **1300**, Show Analysis Process **1400**, and Recommendation Process **1500**.

[0023] In an embodiment of Create User Home Items **500**, User Home Items are created. User Home Items may comprise a selection of a Brand or Merchant (referred to as "User Home **305**") as well as a selection of a competitor of the User Home **305** (referred to as "Competitor **310**"). Generally, a "Brand" may be understood as a trademark under which products or services are sold; products and services are referred to herein as Product(s) **327**. Products **327** are unique products or services made by a manufacturer or provided by a service provider, each separate Product **327** generally being sold under a SKU number, Universal Product Code ("UPC"), product or service name, or the like. Indix Corporation has crawlers, such as Crawl Agent **110**, and data analysis methods and systems which identify Products **327**, Price Attributes **340**, and Product Attributes **345** (discussed at greater length in U.S. application Ser. Nos. 13/951,248 and 13/951,244). In the claims, Price Attributes **340** and Product Attribute **345** are both referred to as "product information". User Home Items are used in other of the processes discussed herein to present information to the user, generally, though not exclusively, relating to the User Home **305** and/or the Competitor **310**.

[0024] In addition to creating User Home Items, Create User Home Items **500** also allows a user to identify Products **327** as Linked Products **315**. When two Products **327** are identified as Linked Products **315**, a search engine routine executed by the Indix Server **200** may increase the frequency with which the Crawl Agent **110** access websites which contain information regarding the Linked Products **315**. Other processes executed by the Indix Server **200**, such as "Substitution Routine **800**" described in U.S. application Ser. No. 13/951,248, may identify the Linked Products **315** as "Substitutes", which, generally, indicates two Products **327** have similar Product Attributes **345** and may be substituted, one for the other.

[0025] In an embodiment of Create List Process **600**, User List(s) **320** may be created. User List(s) **320** may comprise Custom Lists and Smart Lists. User List(s) **320** may comprise a User Home **305**, a Category **335**, a Product **315** and/or a list of Products **315** (not necessarily part of the User Home **305**) which are selected by the user and/or uploaded to the Create List Process **600**.

[0026] In an embodiment of Alert Process **700**, Alert(s) **355** may be created relative to User Home Items and/or User List(s) **320**. The Alert(s) **355** may be triggered, for example, when the relative or absolute price of a Product **327**, a Category **335** of Products **327**, Products **327** at a Merchant or the like changes or passes a threshold.

[0027] In an embodiment of Report Process **800**, Report(s) **350** may be created relative to User Home Items and/or User List **320** (User List **320** is described further herein). Report (s) **350** (discussed further herein) may be setup to trigger actions, such as to re-price or order more of a Product **327**.

[0028] In an embodiment of Summary Process **900**, Insight Information **330** may be provided relative to User Home Items and/or User List **320**. "Insight Information **330**" is the output of analytic processes ("Insights **375**" determined by "Insights Routine **600**" as described in U.S. patent application Ser. No. 13/951,248) which analytic processes identify, for example, price changes, and what Product Attributes **345** and Price Attributes **340** across the datasets are associated with changes in price. Insight Information **330** may comprise the output of analysis directed to determining at least, for example, the following: i) the volatility of prices relative to Attributes **345**, Price Attributes **340**, and other data available to the Indix Server **200**; ii) Substitutes for Products **327** and Categories **335** of Products **327**; iii) the number of Products **327**, Brands, and Categories **335** of Products **327** in the many datasets available to the Indix Server **200**; iv) predicting the future price of Products **327**; v) competitors relative to a Merchant, Store or Brand, vi) promotions of Products, Stores, or Brands; vii) which Products lead or follow others in terms of price changes; viii) which Products in a Category **335** charge higher (premium) prices; ix) the number of price ranges and maximum and minimum for Products and Categories **335** of products; and x) the reach of Products **327** in terms of the number of people who visit a physical or online sales venue. In the claims, Insight Information **330** are part of a group which are referred to as the result of analysis of the set of product information.

[0029] In an embodiment of Quick View Process **1000**, Insight Information **330** may be provided relative to User List **320**, in color coded tiles and with recommendations generated by Recommendation Process **1500**. Quick View Process **1000** may also provide information refining functions to a user.

[0030] In an embodiment of Show Products Process **1100**, information regarding Product(s) **327** may be provided to a user with functions to refine the set of Product(s).

[0031] In an embodiment of Show Promotions Process **1200**, promotion information relative to Products **327** may be provided to a user.

[0032] In an embodiment of Show Social Process **1300**, information regarding social network events relative to User Home Item(s) and/or User List(s) **320** may be provided to a user.

[0033] In an embodiment of Show Analysis Process **1400**, Insight Information **330** may be obtained, patterns therein may be identified, a scale may be selected to fit the observed patterns, the Insight Information **330** may be output in graphical or textual form, user criteria to refine the output may be received and implemented, and the output may be assigned to a Dashboard.

[0034] In an embodiment of Recommendation Process **1500**, two or more Products provided by different Merchants or at different Stores may be compared and, based on the comparison, recommendations may be made to add or remove Products from inventory, to increase or decrease prices, or to increase or decrease Promotions of a Product.

[0035] The phrases "in one embodiment", "in various embodiments", "in some embodiments", and the like are used repeatedly. Such phrases do not necessarily refer to the same embodiment. The terms "comprising", "having", and "including" are synonymous, unless the context dictates otherwise.

[0036] Reference is now made in detail to the description of the embodiments as illustrated in the drawings. While embodiments are described in connection with the drawings and related descriptions, there is no intent to limit the scope to the embodiments disclosed herein. On the contrary, the intent is to cover all alternatives, modifications and equivalents. In alternate embodiments, additional devices, or combinations of illustrated devices, may be added to, or combined, without limiting the scope to the embodiments disclosed herein.

[0037] FIG. **1** illustrates a computing environment **100** configured according to an embodiment. Generally, Indix Server **200** connects to Indix Database **300** and Network **150**. Also connected to Network **150** are Client Device **105**, Crawl Agent **110**, and Ecommerce Platform **115**. Connection to Network **150** or direct connection between computing devices may require that the computers execute software routines which enable, for example, the seven layers of the Open System Interconnection (OSI) model of computer networking or equivalent in a wireless phone or wireless data network. Network **150** comprises computers, network connections among the computers, and software routines to enable communication between the computers over the network connections. Network **150** may comprise, for example, an Ethernet network and/or the Internet. Communication among the various computers and routines may utilize various data transmission standards and protocols such as, for example, the application protocol HTTP. Transmitted data may encode documents, files, and data in various formats such as, for example, HTML, XML, flat files, and JSON. Connection between Indix Server **200** and Indix Database **300** may be via Network **150**.

[0038] FIG. **2** illustrates several components of an exemplary Indix Server **200** in accordance with one embodiment. In various embodiments, Indix Server **200** may include a desktop PC, server, workstation, mobile phone, laptop, tablet, set-top box, appliance, or other computing device that is capable of performing operations such as those described herein. In some embodiments, Indix Server **200** may include many more components than those shown in FIG. **2**. However, it is not necessary that all of these components be shown in order to disclose an illustrative embodiment.

[0039] In various embodiments, Indix Server **200** may comprise one or more physical and/or logical devices that collectively provide the functionalities described herein. In some embodiments, Indix Server **200** may comprise one or more replicated and/or distributed physical or logical devices and/or components.

[0040] In some embodiments, Indix Server **200** may comprise one or more computing resources provisioned from a "cloud computing" provider, for example, Amazon Elastic Compute Cloud ("Amazon EC2"), provided by Amazon.com, Inc. of Seattle, Washington; Sun Cloud Compute Utility, provided by Sun Microsystems, Inc. of Santa Clara, California; Windows Azure, provided by Microsoft Corporation of Redmond, Washington, and the like.

[0041] Indix Server **200** includes bus **220** interconnecting several components including network interface **230**, optional display **240**, optional input **245**, central processing unit **210**, and memory **250**.

[0042] Memory **250** generally comprises a random access memory ("RAM") and permanent non-transitory mass storage device, such as a hard disk drive or solid-state drive. Memory **250** stores program code for the following processes: UI Routine **400** for managing the other processes discussed herein; Create User Home Items **500**; Create List Process **600**; Alert Process **700**; Report Process **800**; Summary Process **900**; Quick View Process **1000**; Show Products Process **1100**; Show Promotions Process **1200**; Show Social Process **1300**; Show Analysis Process **1400**, and Recommendation Process **1500**. In addition, memory **250** also stores operating system **255**.

[0043] These and other software components may be loaded into memory **250** using a drive mechanism (not shown) associated with a non-transitory computer-readable medium **295**, such as a floppy disc, tape, DVD/CD-ROM drive, memory card, or the like.

[0044] Memory **250** also includes Indix Database **300**, illustrated further in FIG. **3**. In some embodiments, Indix Server **200** may communicate with Indix Database **300** via network interface **230**, a storage area network ("SAN"), a high-speed serial bus, and/or via the other suitable communication technology.

[0045] The browser routine referred to above may provide an interface for interacting with other computers through, for example, a webserver and a web browser routine (which may serve and request data and information in the form of webpages). The web browsers and webservers are meant to illustrate or refer to user- and machine-interface and user- and machine-interface enabling routines generally, and may be replaced by equivalent routines for obtaining information from, serving information to, and rendering information in a user or device interface. An application program interface (or "API") may be used to facilitate communication among computers and routines executed by computers. Login credentials and local instances of user or device profiles may be stored in or be accessible to Indix Server **200**, Client Device **105**, Ecommerce Platform **115**, and Crawl Agent **110**. Such user or device profiles may be utilized to provide secure communication between the computers.

[0046] Client Device **105**, Crawl Agent **110** and Ecommerce Platform **115** may be provided by computing devices similar to Indix Server **200**.

[0047] Indix Database **300**, illustrated in FIG. **3**, illustrates data groups used by routines. In addition to the data groups explicitly illustrated, additional data groups may also be present on and/or executed by this device, such as routines for databases, webservers, and web browsers, and routines to enable communication with other computers. The data groups used by routines may be represented by a cell in a column or a value separated from other values in a defined structure in a digital document or file. Though referred to herein as individual records or entries, the records may comprise more than one database entry. The database entries may be, represent, or encode numbers, numerical operators, binary values, logical values, text, string operators, joins, conditional logic, tests, and similar.

[0048] The data groups used by routines and illustrated in FIG. **3** are discussed further herein in relation to the routines which utilize these data groups. A summary of these data groups is also provided, as follows:

[0049] Demand Signal **301** is a record recording demand signals for a Product. Demand Signal **301** is generally intended to comprise signals indicative of demand for a Product, such as whether demand for a Product is increasing or decreasing, whether demand is elastic or inelastic, and the like. Demand Signal **301** may be based on the availability and/or availability delta of a Product, the traffic to websites selling the Product, and the like. Demand Signal **301** may be determined as described in U.S. Provisional Patent Application Ser. No. 61/952,029 and child applications thereof in relation to "Demand Signal Determination Process **500**". In the claims, Demand Signal **201** is part of the group which are referred to as the result of analysis of the set of product information.

[0050] [Para SO] Social Signal **303** is a record recording social media response to a Product, such as a number of "likes", "follows", positive and negative "tweets", "mentions", and the like, which number may be normalized relative to all Products or Products within a Category **335** in which the Product may be found.

[0051] User Home **305** is a record recording a Brand, Merchant, or Store selected by the user, selected according to an employer of the user, or in which the user is otherwise interested. User Home **305** may be created during execution of Create User Home Items **500** routine.

[0052] Competitor **310** is a record of one or more competitors of User Home **305** and may comprise a Brand, Merchant, or Store. Competitor **310** may be selected by a user or may be provided, optionally with confirmation, to the user. Who are competitors of User Home **305** may be determined by, for example, "Competition Routine **1100**" described in application Ser. No. 13/951,248. In brief, with respect to Stores, Brands, and Merchants, Competitors **310** may be determined by identifying Stores, Brands, and Merchants which share a threshold level of the same Products **327**, such as 70%.

[0053] Linked Products **315** is a record recording Products **327** which have been linked pursuant to Create User Home Items **500** routing. Linked Products **315** is described further in relation to FIG. **5**.

[0054] User List **320** is a record recording the result of Create List Process **600**. User List **320** is described further in relation to FIG. **6**.

[0055] Parse Result **325** is a record recording content obtained from a URI by "Crawl Agent **400**" as described in U.S. application Ser. No. 13/951,248 (corresponding to Crawl Agent **110**). Generally, Parse Result **325** (described at greater length in U.S. application Ser. Nos. 13/951,248 and 13/951,244) may comprise a Product name, available in number of Stores, Product graphic, Store, price, promotion status, availability and the like. Information in Parse Result **325** records may be stored in, for example, Price Attribute **340** or Product Attribute **345** records.

[0056] Product **327** (or simply "Product") is a record recording a particular Product and may comprise a Product name, a product identifier (such as a UPC, SKU), and the like. In U.S. application Ser. Nos. 13/951,248 and 13/951,244, "MPID" or "MPID 332" is an identifier generally meant to identify a single Product, generally produced by a common manufacturer, though the Product may be distributed and sold by multiple parties. Product **327** as used herein is similar to "MPID" and MPIDs may be assigned as Products **327**, though Product **327** may also include an identifier, such as a name, trademark, or the like. Product **327** may also be understood as a value within a Product Attribute **340** record, as discussed in application Ser. Nos. 13/951,248 and 13/951,244.

[0057] Insight Information **330** is a record recording the result of analytic processes ("Insights **375**" determined by "Insights Routine **600**" as described in U.S. patent application Ser. No. 13/951,248) which analytic processes identify, for example, price changes, and what Product Attributes **345** and Price Attributes **340** across the datasets are associated with changes in price. Insight Information **330** may comprise the output of analysis directed to determining at least the following: i) the volatility of prices relative to Attributes **345**, Price Attributes **340**, and other data available to the Indix Server **200**; ii) Substitutes for Products **327** and Categories **335** of Products **327**; iii) the number of Products **327**, Brands, and Categories **335** of Products **327** in the many datasets available to the Indix Server **200**; iv) predicting the future price of Products **327**; v) competitors relative to Product, Store, or Brand, vi) promotions of Products, Stores, or Brands; vii) which Products lead or follow others in terms of price changes; viii) which Products in a Category **335** charge higher (premium) prices; ix) the number of price ranges and maximum and minimum for Products and Categories **335** of products; and x) the reach of Products **327** in terms of the number of people who visit a physical or online sales venue. In the claims, Insight Information **330** is part of the group which are referred to as the result of analysis of the set of product information.

[0058] Category **335** is defined further in U.S. application Ser. Nos. 13/951,248 and 13/951,244 and, generally, is an entry in a hierarchical categorization or taxonomy in which Products **327** are arranged. For example, a hierarchical categorization or taxonomy may take the following form: <clothing/shoes/dress shoes/high heels>. In this example, <high heels> may be recorded in a Category **335** record. The record may refer or be locatable in the location in the taxonomy represented by <clothing/shoes/dress shoes/high heels>.

[0059] Price Attributes **340** and Product Attribute **345** records may record information derived from Parse Result **325** records. Price Attribute **340** and Product Attribute **345** records are further defined in U.S. application Ser. Nos. 13/951,248 and 13/951,244. Generally, a Price Attribute **340** record may comprise one or more records recording, for example, a price observed at a particular time, a Product Name (a "Product Name" value in this record may also be referred to herein as "Product" and "Product **327**"), a Product identifier (such as a SKU number), a Standard Price, a Sale, a Price, a Rebate amount, a Price Instructions record (containing special instructions relating to a price, such as that the price only applies to students), a Currency Type, a Date and Time Stamp, a Tax record, a Shipping record (indicating costs relating to shipping to different locations, whether tax is calculated on shipping costs, etc.), a Price Validity Start Date, a Price Validity End Date, a Quantity, a Unit of Measure Type, a Unit of Measure Value, a Merchant Name (with the name of a merchant from whom the Product is available; a "Merchant Name" value in this record may also be referred to herein as a "Merchant"), a Store Name (a Merchant may have multiple stores; a "Store Name" value in this record may also be referred to herein as a "Store"), a User ID, a Data Channel (indicating the source of Price Attribute **340** record, such as an online crawl, a crowd-source, a licensed supplier of price information, or from a merchant), a Source Details record (for example, indicating a URI, a newspaper advertisement), an Availability Flag, a Promotion Code, a Bundle Details record (indicating products which are part of a bundle), a Condition Type record (indicating new, used, poor, good, and similar), a Social Rank record (indicating a rank of "likes", "followers" and similar of the price), a Votes/Likes record (indicating a number of "likes", "follows" and similar which a Price or Product has received), a Price Rank record, a Visibility Indicator record (indicating whether the price is visible to the public, whether it is only visible to a Merchant, or the like), a Supply Chain Reference record (indicating whether the price was obtained from a retailer, a wholesaler, or another party in a supply chain), a Sale Location (indicating a geographic location where the product is available at the price), a Manufactured Location record (indicating where the product was produced or manufactured), a Launch Date record (indicating how long the product has been on the market), and an Age of Product record (indicating how long the product was used by the user). When capitalized herein, the foregoing terms (such as Product, Price, Merchant, Store, Source Details, etc.) are meant to refer to values in a Price Attribute **340** or Product Attribute **345** record.

[0060] A Product Attribute **345** record may comprise, for example, values encoding features of or describing a Product. The entire Product Attribute **345** schema may comprise thousands of columns, though only tens or hundreds of the columns may be applicable to any given Product. An example set of values in a Product Attribute **345** record for a ring is as follows: Title, "Sterling Silver Diamond & Blue Topaz Ring;" Brand, "Blue Nile;" Category (such as, for example, a Category **335** in a category schema), "rings;" Metal Name, "silver;" Stone Shape, "cushion;" Stone Name, "topaz;" Width, "3 mm;" Stone Color, "blue;" Product Type, "rings," Birthstone, "September;" and Setting Type, "prong." An example set of Product Attributes **345** for a shoe is as follows: Brand, "Asics;" Category (such as, for example, a Category **335** in a category schema or taxonomy), "Men's Sneakers & Athletic;" Shoe Size, "8;" Product Type, "wrestling shoes," Color, "black;" Shoe Style, "sneakers;" Sports, "athletic;" Upper Material, "mesh." When capitalized herein, the foregoing terms (such as Brand, Category, Metal Name, Product Type, etc.) are meant to refer to values in a Product Attribute **345** record.

[0061] Report **350** record may be produced during execution of Report Process **800**. Report **350** record is discussed further in relation to FIG. **8**.

[0062] Alert **355** record may be produced during execution of Alert Process **700**. Alert **355** record is discussed further in relation to FIG. **7**.

[0063] User Tag **380** record records an identifier assigned to a User List **320** by a user pursuant to Create List Process **600**. User Tag **380** is discussed further in relation to FIG. **6**.

[0064] Promotions **382** record records the number, length, date/time, and magnitude of promotions (discounts) of a Product, either as advertised or relative to an external criteria, such as an average or mean price. Information in Promotions **382** record may be obtained by, for example, "Promotion Routine **1200**" described in U.S. patent application Ser. No. 13/951,248, and as discussed in U.S. Provisional Patent Application Ser. No. 61/952,029 and child application(s) thereof in relation to "Promotion **382**" record therein. In the claims, Promotions **382** is part of the group which are referred to as the result of analysis of the set of product information.

[0065] User Analysis **385** record records a user-customized selection of Insight Information **330**, pursuant to Show Analysis Process **1400**. User Analysis **385** is discussed further in relation to FIG. **14**.

[0066] FIG. **4** illustrates an embodiment of User Interface Process **400**.

[0067] At block **405**, Client Device **105** may log into, for example, User Interface Process **400**. Any of processes **500** through **1400** may be executed independently by User Interface Process **400**, not necessarily in the order presented in FIG. **4**. Processes **500** through **1500** are discussed at great length herein.

[0068] Briefly, at block **500** in FIG. **4**, Create User Home Items **500** may be executed. Create User Home Items **500** may receive, for example, user selection of User Home **305**. User Home **305** may be a Brand, Merchant, or Store selected by the user, selected according to an employer of the user, or in which the user is otherwise interested. Create User Home Items **500** may also receive identification and/or selection of Competitor **310**, Products **327** of interest to the user, and user designation of link(s) between Products **327**. Create User Home Items **500** may also set two or more Products **327** as "Substitutes" if the number of user designated links between the Products **327** exceeds a threshold. Create User Home Items **500** is described further in relation to FIG. **5**.

[0069] At block **600** in FIG. **4**, Create List Process **600** may be executed. Create List Process **600** may receive a list of Products **327**—*a* "custom list"—and/or criteria for including Products **327** in a list-a "smart list". The criteria for including Products **327** in a list may comprise a Category, a Brand, a Store or the like. Create List Process **600** is described further in relation to FIG. **6**.

[0070] At block **700** in FIG. **4**, Alert Process **700** may be executed. Alert Process **700** is a process wherein users can set alerts to occur when one or more targets (such as a Product **327** or a Category **335** of Products **327**) obtains a specified criteria, such as an absolute or relative change in price, or price ratio relative to another target, or changes availability at a Store (such as becomes available or is no longer available). Alert Process **700** is described further in relation to FIG. **7**.

[0071] At block **800** in FIG. **4**, Report Process **800** may be executed. Repot Process **800** is a process wherein users can create customized reports and set re-pricing commands. Report Process **800** is described further in relation to FIG. **8**.

[0072] At block **900** in FIG. **4**, Summary Process **900** may be executed. Summary Process **900** is a process wherein summaries of Insight Information **330** records may be customized for and presented to a user. Summary Process **900** is described further in relation to FIG. **9**.

[0073] At block **1000** in FIG. **4**, Quick View Process **1000** may be executed. Quick View Process **1000** is a process wherein a user may obtain multiple types of Insight Information **330** records and output them with color. Quick View Process **1000** is described further in relation to FIG. **10**.

[0074] At block **1100** in FIG. **4**, Show Products Process **1100** may be executed. Show Products Process **1100** is a process wherein a user may obtain, refine, and sort information relating to User Home Items and User Lists, as described further in relation to FIG. **11**.

[0075] At block **1200** in FIG. **4**, Show Promotions Process **1200** may be executed. Show Promotions Process **1200** is a process wherein information regarding Products **327** on sale may be output to a user. Show Promotions Process **1200** is described further in relation to FIG. **12**.

[0076] At block **1300** in FIG. **4**, Show Social Process **1300** may be executed. Show Social Process **1300** is a process wherein information regarding social network events may be output to a user. Show Social Process **1300** is described further in relation to FIG. **13**.

[0077] At block **1400** in FIG. **4**, Show Analysis Process **1400** may be executed. Show Analysis Process **1400** is a process wherein Insight Information **330** relating to a Brand, Category, User List or the like may be adjusted, refined, and assigned to a user dashboard. Show Analysis Process **1400** is described further in relation to FIG. **14**.

[0078] Occurring within one or more of the routines of User Interface Process **400** (as discussed in relation to such routines, herein), Recommendation Process **1500** may be executed to produce recommendations to increase or decrease a price, to add or remove a Product from inventory, or to increase or decrease promotion of a Product.

[0079] FIG. **5** illustrates an embodiment of Create User Home Items **500** routine. At block **505**, User Home **305**—such as a Brand, Merchant, and/or Store with which the user is associated or in which the user is interested-is received, such as from the user via Client Device **105**, and stored. If the user is employed by a corporation or another organization, and if the corporation is obtaining services from Indix Server **200**, User Home **305** may be set by the employer. At block **510**, an identifier may be assigned to the User Home **305** of block **505**, which identifier may be recorded in a User Home **305** record associated with the user.

[0080] At block **515**, competitors of User Home **305** may be received from the user, from a service, from an employer, and/or may be assigned by Create User Home Items **500** process, such as according to Competitor **310** record. Competitor **310** may be determined by, for example, "Competition Routine **1100**" described in U.S. patent application Ser. No. 13/951,248. In brief, with respect to Stores, Brands, and Merchants, Competitors **310** may be determined by identifying Stores, Brands, and Merchants which share a threshold level of the same Products **327**, such as 70%. Create User Home Items **500** process may receive an assigned Competitor **310** from the user or Create user Home Items **500** may suggest or assign competitors of User Home **305**, which suggested or assigned competitors may be confirmed by the user. Create User Home Items **500** may show the user the competitors which other users have selected in relation to User Home **305** selected or assigned in block **505** and may allow the user to select from or approve such competitors. At block **520**, Competitor **310** identifier may be assigned to or associated with the competitors of User Home **305** received at block **515**.

[0081] At block **525**, first and second Products **327** may be received. At block **530**, a designation may be received to link the first and second Products **327** of block **525**. At block **535**, the first and second Products **327** of block **525** may be linked as Linked Products **315**. At block **540**, crawling performed by the Crawl Agent **110** may be increased (such as to the maximum) for one or both of the Linked Products **315** with respect to URI's associated with the Linked Products **315**. The increase may have a temporal effect on the behavior of the Crawl Agent **110** and may be influenced by the frequency with which the first and second Products **327** of block **525** are also Linked Products **315** by the same or different users.

[0082] At block **545**, the link count-which measures the frequency with which the first and second Products **327** of block **525** are also Linked Products **315** by the same or different users-is obtained.

[0083] At block **550**, a determination may be made regarding whether the link count is above a threshold. If so, then at block **555**, the Linked Products **315** may be set as "Substitutes" in, for example, "Insights Routine **700**" described in U.S. patent application Ser. No. 13/951,248. If not, then at block **560**, the below-threshold link count may be ignored.

[0084] At block **599**, Create User Home Items **500** ends or returns to FIG. **4**, block **500**, or to another process which may have called Create User Home Items **500**.

[0085] FIG. **6** illustrates an embodiment of Create List Process **600**.

[0086] At block **605**, a create list command may be received from a user or another process. The create list command may create a User List **320** record. At block **610**, a determination may be made (or equivalent logic implemented) regarding whether the create list command is in relation to a custom list or a smart list; a custom list may be a list of Products **327** or another list uploaded by the user, an employer, or the like while a smart list may be a selection by a user, employer, or the like of a User Home **305** or a Category **335**.

[0087] If, at block **610** the determination was a custom list, then at block **615** the user's product selection for inclusion in the custom list may be received. For example, Products **327** may be selected by the user (or an employer) from a list or catalog, via search results, via browsing a catalog, and the like; the product selection may be in relation to a User Home **305** or a Competitor **310**, such as Products **327** associated with the user's User Home **305** or Competitor **310**; the product selection may further be in relation to a list of products uploaded to the Create List Process **600** by the user. In this and other steps discussed herein, when a user uploads a list of products to a process, the uploaded list may be normalized relative to the Products **327** maintained in the Indix Database **300**, which normalization may involve interaction with the user to clarify any ambiguous or undeterminable items, with the output thereof being used in the relevant process.

[0088] At block **620** the selected or identified products may be saved as a draft, which may allow the list to be refined by the user at a later date. At block **625** a submit command or equivalent may be received from the user and, at block **630**, the product list may be saved as or in association with a User List **320** record.

[0089] If, at block **610** the determination was that a smart list was to be created, then at block **635** a determination may be made regarding whether the smart list is in relation to a Brand or Merchant (or Store) or a Category **335**. The Brand or Merchant may be a User Home **305**, User Home Item or a selection of a Brand or Merchant from a navigable list presented to the user for this purpose. An example Category **335** record may be as follows: apparel/shoes/women/high heel.

[0090] If, at block **635**, the determination was of a category, then at block **640** the user's selection of a Category **335** may be received. At block **645** additional criteria which may be used to limit or refine Category **335** or block **640** may be received. For example, if the user selected a "high heel" Category **335**, the user may further be able to refine the list according to a Brand within this Category **335** or all items in the Category **335** over a certain price level (such as all high heels over $50), or available by or from a certain Brand or Merchant, or at a Store, or within a sub-category with Category **335**.

[0091] If, at block **635** the determination was made to select a Brand or Merchant, then at block **650** the selected Brand or Merchant (which may be a User Home **305**) may be received. At block **655** the user's criteria within the Brand or Merchant may be received. For example, if the user selected a Merchant, the user may further be able to refine the list according to a Brand sold by the Merchant, Brands at a particular Store of the Merchant, or Products over a certain price level; if the user selected a Brand, the user may further be able to refine the list according to specific products within the Brand or products within the Brand over a certain price level.

[0092] At block **660**, the selected and refined Brand or Merchant information may be saved as a smart list.

[0093] At block **665**, a user tag, name, or identifier may be received from the User, saved as User Tag **380**, and assigned to User List **320**. User Tag **380** may be used to identify the User List in other routines.

[0094] At block **699**, Create List Process **600** ends or returns to FIG. **4**, block **600**, or to another process which may have called Create List Process **600**.

[0095] FIG. **7** illustrates an embodiment of Alert Process **700**.

[0096] At block **705**, a create alert command may be received from a user or another process. At block **710**, an alert type selection may be received. The alert types may be, for example, an alert based on a price change of a Product **327**, Products in a Category **335**, a User Home **305**, a Competitor **310**, a User List **320**, based on a price comparison between two or more Products **327**, Categories **335**, User Home **305**, Competitor **310**, Products in a User List **320**, based on a promotion or withdrawal of promotion of (such as advertised or actual percentage change in price) a Product **327**, User Home **305**, Competitor **310**, or a User List **320**, and the like. The alert type may be responsive to, for example, the output of Recommendation Process **1500** and a recommendation of a type or with a specified value.

[0097] At block **715**, Alert Process **700** may receive and save the first target of the alert command, such as a Product **327**, User Home **305**, Competitor **310**, User List **320**, or a user selection of a Brand, Merchant, or Category **335**.

[0098] At block **720** the criteria for the alert may be received and saved, such as whether the alert will be triggered upon an increase or decrease of a price of the first target, either absolutely relative to an earlier value, as a percentage relative to an earlier value, or absolutely or as a percentage relative to a second target, whether a product has gone on a promotion, has been discontinued, is not available, or based on a defined recommendation being generated by Recommendation Process **1500**, or the like.

[0099] At block **725**, the second target, if any (as in the case of a comparison between products), may be received and saved.

[0100] At block **730**, the notification window, frequency, and other notification criteria of the alert may be received and saved.

[0101] At block **735**, the alert may be executed or may be set to be executed and notifications resulting therefrom may be sent.

[0102] At block **799** Alert Process **700** ends or returns to FIG. **4** and block **700** or to another process which may have called Alert Process **700**.

[0103] FIG. **8** illustrates an embodiment of Report Process **800**.

[0104] At block **801**, a create report command may be received from a user or another process. At block **805**, a report type selection may be received from a user or another process. The report types may be, for example, a "MAP" report in relation to products that are priced below a minimum advertised price, a pricing opportunity report in relation to products that are priced above or below a competitor's price, or a product availability report in relation to products that are out of stock. The report type may be based on a result of Recommendation Process **1500**.

[0105] At block **810**, a data selection for the report may be received from a user or another process. The data selection may be a User List **320**, such as a Custom or Smart List, a selection of a Brand, Category **335**, or Merchant, Store, an uploaded list, or the like.

[0106] At block **815**, data refinement selections may be received from a user or another process. The data refinements may comprise criteria such as, for example, "where the price of X product at Y Store is higher/lower than the minimum or maximum of a price for X product at Competitor **310** by at least a specified amount."

[0107] At block **817**, report column configuration settings may be received, such as to include or not include in the report a product title, a product SKU, a Brand name, an MPN, a UPC, a difference from lowest competitor, a "my sale price", a sale price across competitors, and the like.

[0108] At block **820** a determination may be made regarding whether a re-price command is received or programmed. A re-price command may be an instruction to re-price, add, or remove a product in response to conditions or information in a report. If so, then at block **825**, a user instruction to match, price higher than or price lower than may be received. At block **830**, a user instruction may be received that the re-price instruction is relative to a minimum or maximum of a competitor's price or a sale price at a selected competitor.

[0109] At block **835**, delivery options for the report may be received from a user or another process. The delivery options may be, for example, whether the report is to be emailed when ready (which may include emailing a spreadsheet or a link to a spreadsheet or the like) or provided on a schedule, or the like.

[0110] At block **840**, the report may be saved as or in, for example, a Report **350** record.

[0111] At block **845**, the report may be executed or may be scheduled to be executed and the result thereof delivered.

[0112] At block **899**, Report Process **800** ends or returns to FIG. **4** and block **800** or to another process which may have called Report Process **800**.

[0113] FIG. **9** illustrates an embodiment of Summary Process **900**.

[0114] At block **905**, a summary command may be received from a user or another process. The summary may be a default setting of the user interface.

[0115] Beginning in opening loop block **910**, Summary Process **900** processes each User Home Item, User List **320**, favorite or other user selection in turn.

[0116] At block **915**, Insight Information **330** for each item of block **910** may be obtained. Insight Information **330** is the output of analytic processes ("Insights **375**" determined by "Insights Routine **600**" as described in U.S. patent application Ser. No. 13/951,248) which analytic processes identify, for example, price changes, and what Product Attributes **345** and Price Attributes **340** across the datasets are associated with changes in price. Insight Information **330** may comprise, for example, a number of products, number of categories or subcategories, number of Brands, number of Merchants, number of Stores, number of price increases or decreases, number added in the last week or other date-time unit, number out of stock, number on sale, or the like.

[0117] At block **920**, tiles comprising a numeric value of the Insight Information **330** of block **915** may be output. At block **925**, graphs comprising a graphical representation of the Insight Information **330** of block **915** may be output.

[0118] In ending loop block **930**, Summary Process **900** iterates back to opening loop block **910** to process the next User Home Item, User List **320**, favorite or other user selection, if any.

[0119] At block **999** Summary Process **900** ends or returns to FIG. **4** and block **900** or to another process which may have called Summary Process **900**.

[0120] FIG. **10** illustrates an embodiment of Quick View Process **1000**.

[0121] At block **1005**, a quick view command may be received from a user or another process. Beginning in opening loop block **1010**, Quick View Process **1000** processes each User Home Item, User List **320**, favorite or other user selection in turn.

[0122] At block **1015**, Insight Information **330** for each item of block **1010** may be obtained. The Insight Information **330** is described in relation to block **915** and elsewhere and may comprise, for example, a number of products, number of categories or subcategories, number of Brands, number of Stores, number of price increases or decreases, number added in the last week or other date-time unit, number out of stock, number on sale, or the like.

[0123] At block **1020**, price-related Insight Information **330** may be output in association with a color A.

[0124] At block **1025**, assortment-related Insight Information **330** may be output in association with a color B.

[0125] At block **1030**, promotion-related Insight Information **330** may be output in association with a color C.

[0126] At block **1035**, Linked Products **315** may be output in association with a color D.

[0127] In ending loop block **1040**, Quick View Process **1000** iterates back to opening loop block **1010** to process the next item, if any.

[0128] At block **1500**, recommendations regarding whether Products may be added or removed from inventory, or whether the price of a Product may be increased or decreased, or whether a Product should be promoted more or less may be obtained by executing Recommendation Process **1500**.

[0129] At block **1042**, the recommendations generated at block **1500** may be output, for example, to Client Device **105**.

[0130] At block **1045**, a user selection or deselection of an output detail of blocks **1010** to **1040** and of **1042** may be received. The user selection or deselection may, for example, allow the output detail to be shown or not shown.

[0131] At block **1050**, the output may be updated to reflect the selection or deselection of block **1050**.

[0132] At block **1099** Quick View Process **1000** ends or returns to FIG. **4** and block **1000** or to another process which may have called Quick View Process **1000**.

[0133] FIG. **11** illustrates an embodiment of Show Products Process **1100**.

[0134] At block **1105**, a show product information command may be received from a user or another process.

[0135] Beginning in opening loop block **1110**, Show Products Process **1100** processes each User Home Item, User List **320**, favorite or other user selection in turn.

[0136] At block **1115**, the product name, available in number of Stores, product graphic, Store, price, promotion status, availability and the like information may be obtained, such as from Price Attribute **340**, Product Attribute **345**, or Parse Result **325** records (described at greater length in U.S. patent application Ser. Nos. 13/951,248 and 13/951,244). At block **1120**, the information of block **1115** may be output, for example, to Client Device **105**.

[0137] At block **1130**, user sort criteria may be received, such as sort-by relevance, price, name, date, or the like.

[0138] At block **1135**, the output may be updated according to the user sort criteria.

[0139] In ending loop block **1140**, Show Products Process **1100** iterates back to opening loop block **1110** to process the next item, if any.

[0140] At block **1199** Show Products Process **1100** ends or returns to FIG. **4** and block **1100** or to another process which may have called Show Products Process **1100**.

[0141] FIG. **12** illustrates an embodiment of Show Promotions Process **1200**.

[0142] At block **1205**, a show promotion information command may be received from a user or another process.

[0143] Beginning in opening loop block **1210**, Show Promotions Process **1200** processes each User Home Item, User List **320**, favorite or other user selection in turn.

[0144] At block **1215**, Show Promotions Process **1200** gets the products which are on sale, as may be advertised by the merchant or Store or as may be determined relative to a standard or average retail price. Determination of whether a product is on promotion may be performed, for example, by a "Promotions Insight", as described in U.S. patent application Ser. No. 13/951,248, for example, in relation to "Promotion Routine **1200**" therein). At block **1220**, the Show Promotions Process **1200** gets the products which are on sale by various percentage amounts (such as 10%, 20%, 30%, etc.), as may be advertised by the merchant or Store or as may be determined relative to a standard or average retail price.

[0145] At block **1225**, the products which are on sale may be output, such as to the Client Device **105**. At block **1230**, the products which are on sale by various percentage amounts may be output. The output may comprise a line graph of zero to one-hundred with circles at each of 10-unit increments, with the size of the circle determined by the number of products within that sale percentage band.

[0146] In ending loop block **1235**, Show Promotions Process **1200** iterates back to opening loop block **1210** to process the next item, if any.

[0147] At block **1299** Show Promotions Process **1200** ends or returns to FIG. **4** and block **1200** or to another process which may have called Show Promotions Process **1200**.

[0148] FIG. **13** illustrates an embodiment of Show Social Process **1300**.

[0149] At block **1305**, a show social information command may be received from a user or another process.

[0150] Beginning in opening loop block **1310**, Show Social Process **1300** processes each User Home Item, User List **320**, favorite or other user selection in turn.

[0151] At block **1315**, Show Social Process **1300** may obtain the total number and type of social network events associated with the item of block **1310**. This may include, for example, a number of "friends", "likes", "shares", "follows", "tweets", messages mentioning the item, or the like. This information may be normalized relative to all of the items of block **1310** or relative to such events occurring with respect to a Category, such as a Category **335** of an item of block **1310** or a Category **335** selected by the user. This may be referred to in the claims as a "social metric". As an alternative, Social Signal **303** with respect to the items of block may be obtained. Social Signal **303** may be determined as described in U.S. Provisional Patent Application Ser. No. 61/952,029 and child application(s) thereof, in relation to "Social Signal Determination Process **700**", described therein. At block **1320**, the information of block **1315** may be output, such as to Client Device **105**.

[0152] In ending loop block **1325**, Show Social Process **1300** iterates back to opening loop block **1310** to process the next item, if any.

[0153] At block **1399** Show Social Process **1300** ends or returns to FIG. **4** and block **1300** or to another process which may have called Show Social Process **1300**.

[0154] FIG. **14** illustrates an embodiment of Show Analysis Process **1400**.

[0155] At block **1405**, a show analysis center command may be received from a user or from another process.

[0156] At block **1410**, a user selection of a Brand, Merchant, Store, User Home **305**, User Home Item, a Category **335**, a User List **320**, or a quick view may be received. More than one such selection may be received, each of which may be added to a separate tab in the UI for Show Analysis Process **1400**. At block **1415**, the selection(s) may be saved.

[0157] At block **1420**, one or more sets of Insight Information **330** relating to the user selection(s) of block **1410** may be obtained. Insight Information **330** and examples thereof are discussed above, in relation to Summary Process **900**. Which Insight Information **330** to obtain may be a default setting or may be specified by the user. In addition, or instead, signals may be obtained, such as Demand Signal **301**, Supply Signal **302**, Social Signal **303**, Quality Signal **304**, and Product-Type Quantity Signal **306**. These Signals may be determined as described in U.S. Provisional Patent Application Ser. No. 61/952,029 and child applications thereof, such as in relation to FIGS. **5-11** therein. These Signals may also be referred to in the claims as "metrics", such as "demand metric", "social metric", etc.

[0158] At block **1425**, the scale of the graph for Insight Information **330** and signal information of block **1420** may be selected, such as to present the information in a compact form without excessive blank area. The information may then be output, such as to the Client Device **105**, as one or more graphs. The output may allow the user to select an item of block **1415** to be displayed or not displayed in the graph (which may cause the graph to be re-sized).

[0159] At block **1426**, patterns may be identified in the Insight Information **330** or signal information for the user selection(s) of block **1410**, such as repeating patterns and/or patterns where a first pattern with respect to a first Insight Information **330** or first Signal is preceded or followed by a second pattern in a second Insight Information **330** or second Signal. Patterns which occur across more than one Insight Information **330** or Signal may require that the more than one Insight Information **330** or Signal share a common x-axis, such as that they each are part of a time-series. Pattern identification may be according to a continuous probability distribution function or another statistical analysis. Where no pattern can be identified with respect to either a single Insight Information **330**/Signal or between more than one Insight Information **330**/Signal, the pattern detection may return no result or may return a message indicating that no pattern was detected.

[0160] For example, at block **1410**, a user may select a price of a Category **335**, such as "impact wrenches" across all Merchants or with respect to a single Merchant, and a Social Signal with respect to the same Category **335** at the Merchant(s). Pattern identification may reveal that there are repeating patterns in both data sets (in the price data and in the social metric). Pattern identification may further reveal that there is a pattern between the repeating patterns in both data sets, such as that when the price for the Category "impact wrenches" is high, there is also a higher Social Signal **303**. Pattern identification may reveal that the higher Social Signal **303** follows the increase in price.

[0161] At block **1427**, the scale of the Insight Information **330** may be adjusted to adequately encompass patterns identified in block **1421**, for example, such that the units cover the entire range, without excess empty space, and the result may be output, such as to Client Device **105**.

[0162] At block **1430**, the Products **327** in the user selection of block **1410** may be output to the user or another process. These may be output under a separate tab or the like.

[0163] At block **1435**, user refinement of the display may be received. User refinement may comprise limiting the output to specific Brands, Merchants, Stores, Products **327**, price ranges, and the like.

[0164] At block **1440**, the display at the Client Device **105** may be updated to reflect the refinement of block **1435**.

[0165] At block **1445**, the output of block **1440** may be saved as a User Analysis **385** record. This may be in response to a command received from the user or a process.

[0166] At block **1450**, a command may be received from the user or a process to assign the User Analysis **385** record of block **1445** to the "Dashboard" of the user. At block **1455**, the item may be assigned to the user's Dashboard. As time unfolds, the data underlying the output of block **1440** may be updated and, as the user visits the user's Dashboard, the updated data may be visible.

[0167] Throughout this paper, items, not just those in the Analysis Center, may be assigned to a user Dashboard, for quick access to data relating to the updated version of the item. A user Dashboard may be an element in a user interface which quickly and/or persistently presents information.

[0168] At block **1499** Show Analysis Process **1400** ends or returns to FIG. **4** and block **1400** or to another process which may have called Show Analysis Process **1400**.

[0169] FIG. **15** illustrates an embodiment of Recommendation Process **1500**. In discussion of this process, parties "A" and "B" are discussed. A user in a given interaction may be represented by one of these parties, in which case a recommendation to the other party may not be output. For example, if a user in an interaction is represented by "Party A", then a recommendation made (or proposed to be made) to "Party B" may not be output to the user (Party A).

[0170] Blocks **1505** to **1575** may iterate for Product pairs such as a selected i) User Home Item (comprising User Home **305** and Competitor **310**) or ii) a User List comprising at least two different Merchants, Stores, Brands, or Products.

[0171] At block **1510**, a determination may be made regarding whether a Product is found in both of the entities; for example, in both User Home **305** and Competitor **310** or in both of the entries in the User List.

[0172] If affirmative at block **1510**, then Recommendation Process **1500** may proceed to block **1515**, where a determination may be made regarding whether Demand Signal **301** is different between the two items (or groups of items) of blocks **1505** to **1575**. Demand Signal **301** may be generated as described in U.S. Provisional Patent Application Ser. No. 61/952,029 and child applications thereof. Generally, Demand Signal **301** encodes the demand for a Product or set of Products in a Brand or sold by a Store, Merchant, or the like. Determination of a difference may be based on a statistically significant difference, such as two standard deviations in a probability distribution function, or a difference above a threshold.

[0173] If there is not a difference at block **1515**, and if Demand Signal **301** is low at both of the items (or groups of items) of blocks **1505** to **1575**, then Recommendation Process **1500** may recommend that the items be removed.

[0174] If there is a difference at block **1515**, then at block **1525**, a determination may be made regarding whether or not there is a price difference between the items (or groups of items). For example, a price difference may be found when there is greater than two-standard deviations in a continuous probability distribution function. If affirmative at block **1525**, then at block **1530**, a determination may be made regarding whether or not there is difference in promotions between the items (or groups of items). The number of promotions, the magnitude of promotions, and the duration of promotions may be generated as described in U.S.

patent application Ser. No. 13/951,248 in relation to "Promotion Routine **1200**" described therein, and may use data in Promotions **382** record.

[0175] If affirmative at block **1530**, then blocks **1545-1555** describe a matrix. At block **1545**, if Party A has greater demand (according to Demand Signal **301**), lower promotion, and a lower price, then Recommendation Process **1500** may recommend to Party A that Party A has an opportunity to increase price and may recommend to Party B that Party Blower the price and/or increase promotion of the item. At block **1550**, if Party A has greater demand, higher promotion, and a lower price, then Recommendation Process **1500** may recommend to Party A that Party A has an opportunity to increase price and/or decrease promotion and may recommend to Party B that Party Blower the price and/or increase promotion of the item. At block **1555**, if Party A has greater demand, lower promotion, and a higher price, then Recommendation Process **1500** may recommend to Party A that Party A has an opportunity to increase price and may recommend to Party B that Party B remove the item from inventory.

[0176] If negative at block **1530**, then blocks **1535** and **1540** describe a matrix. At block **1535**, if Party A has greater demand (according to Demand Signal **301**) and a lower price, then Recommendation Process **1500** may recommend to Party A that Party A has an opportunity to increase price and may recommend to Party B that Party B increase promotion of the item. At block **1540**, if Party A has greater demand and a higher price, then Recommendation Process **1500** may recommend to Party A that Party A has an opportunity to increase price and may recommend to Party B that Party B increase promotion of the item or remove the item from inventory.

[0177] If negative at block **1510**, then (referring to FIG. **15**B), at block **1560** a determination may be made regarding whether or not the demand for the item (or items) is high, such as according to Demand Signal **301**. If affirmative at block **1560**, then at block **1570**, Recommendation Process **1500** may recommend to Party B that Party B add the item to items offered by Party B. If negative at block **1560**, then at block **1565**, Recommendation Process **1500** may recommend to Party B that Party B remove the item or increase promotion of it.

[0178] At block **1575**, Recommendation Process **1500** may return to block **1505** to iterate over the next Product in the User Home **305** and a corresponding Competitor **310** or the next pair in the User List(s).

[0179] At block **1599** Recommendation Process **1500** may return to Quick View Process **1000** or to another routine which may have called Recommendation Process **1500**.

[0180] Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that alternate and/or equivalent implementations may be substituted for the specific embodiments shown and described without departing from the scope of the present disclosure. This application is intended to cover any adaptations or variations of the embodiments discussed herein.

1. A system comprising a computer processor and a computer memory, the computer memory comprising instructions which, when executed, cause the system to perform operations, in which the operations include:

receiving, via a user computer, over a network, a first user selection including a product, and a brand, vendor and category associated with the product;

accessing a set of product information that includes information regarding the selected product, the set of product information including a set of products, wherein the product information includes for each product in the set: one or more brands under which the product is sold, a set of vendors offering it for sale, and a product category;

generating a set of analysis results based on the obtained product information;

identifying a first statistical pattern in a first analysis result from the set of analysis results for the selected product by performing statistical analysis on the product information and the analysis results,

in which the first statistical pattern is a statistical pattern between two or more types of data included in the first analysis result or in the product information and in which the two or more types of data share a common aspect, and in which the first statistical pattern comprises a cyclic change in the first analysis result,

in which the set of analysis results includes:

a social metric, an identification of which product in a sub-set of products leads or follows other products in the sub-set of products in terms of price changes, a demand metric based at least in part on visitors record generated from webpage traffic to one or more online stores at which the product is available, and in which the demand metric is stored in the database coupled to the computer, and a reach of the product in terms of the number of people who visit an online sales venue of the product, in which the social metric is generated based on a number of followers or a number of likes of the product, and in which the social metric is stored in a database coupled to the computer;

transmitting to the user computer, over the network, for display to the user, the first analysis result and the first statistical pattern;

receiving, via the user computer, over the network, a create alert command for the product, the create alert command including:

alert criteria, including at least one of: absolute or percentage change in price, initiation or termination of sales at a venue, and

a notification window and frequency;

in response to the command, executing the alert and transmit to the user computer, over the network, a notification;

receiving a second user selection and a second analysis result with respect to the second user selection; and

identifying a second statistical pattern in the second analysis result.

2. The system of claim **1**, in which the first user selection further comprises at least one of a user list and a user favorite.

3. The system of claim **2**, in which the user list comprises a custom list or a smart list.

4. The system of claim **3**, in which the smart list is created upon receipt of a selection of at least one of a brand, a vendor or a category in the set of product information and at least one criterion for limiting the first analysis result presented in relation to the user list, and in which the custom list comprises a list of products provided by the user.

5. The system of claim 1, in which the operations further include:

comparing the second analysis result relating to the second user selection to the first analysis result, and

making a first recommendation to the user based on the comparison of the first and second analysis results.

6. The system of claim 5, in which:

the first analysis result comprises, for a product sold by a first vendor: a demand metric for the product sold by the first vendor, a price history for the product sold by the first vendor, and a promotion metric for the product sold by the first vendor;

the second analysis result comprises, for a product sold by a second vendor, a demand metric for the product sold by the second vendor, a price history for the product sold by the second vendor, and a promotion metric for the product sold by the first vendor; and

in which the first recommendation comprises at least one of:

a recommendation to add or remove the product sold by the first vendor or the product sold by the second vendor from an inventory of the first or second vendor,

a recommendation to the first or second vendor to charge a higher or lower price for the product sold by the first vendor or the product sold by the second vendor, and a recommendation to the first or second vendor to increase or decrease a promotion of the product sold by the first vendor or the product sold by the second vendor.

7. The system of claim 6 further including:

determining if the demand metrics in the first and second analysis results are low;

and in response to a determination that they are low, recommending to the user that the product sold by the first vendor or the product sold by the second vendor be removed from inventory.

8. The system of claim 6, further including:

when:

the price histories of the product sold by the first and second vendors indicate a price difference that exceeds a pre-defined threshold, and

there is a difference in the promotion metrics between the product sold by the first and second vendors, at least one of:

recommending an increase in the price of the product sold by the first vendor or the product sold by the second vendor to a vendor associated with a higher demand metric, a lower promotion metric and a lower current price,

recommending an increase in the price of the product sold by the first vendor or the product sold by the second vendor or a lower promotion to a vendor associated with the higher demand metric, a higher promotion metric, and the lower current price, and

recommending that the product sold by the first vendor or the product sold by the second vendor be discontinued by a vendor associated with a lower demand metric, the lower promotion metric, and the lower current price.

9. The system of claim 1, further including:

receiving from the user, over the network, an instruction to link two or more different products in the set of product information; and

in response to a number of times the two or more different products in the set of product information are linked by any user exceeding a threshold, identifying the two or more products as substitutes for each other.

10. The system of claim 1, further including:

receiving an instruction to send a notice to the user when the first analysis result corresponds to a user-specified alarm setting.

11. The system of claim 10, in which the user-specified alarm setting comprises at least a first target in the first user selection and one or more alarm criteria.

12. The system of claim 11, in which the one or more alarm criteria comprise at least one of:

an absolute or percentage change in price of the first target,

a promotion of the first target, and

an inception of availability or discontinuation of availability of the first target.

13. The system of claim 11,

in which the one or more alarm criteria comprise a second target comprising at least one of: a product, a category in a product categorization schema, a brand, and a vendor, and

in which the one or more alarm criteria comprise at least one of an absolute or percentage change in price between the first and second targets and a change in the promotion status of one of the first and second targets relative to either the first or the second target.

14. The system of claim 1, in which the operations further include:

associating the user with a competitor of the user selection and in which presenting the analysis result comprises presenting the first analysis result in relation to the user selection and the competitor.

15. The system of claim 1, in which the operations further include:

receiving a re-price command from the user, and, in response, reprice a product offered by the user in response to a condition of the first analysis result, and

in which the re-price command comprises at least one of an instruction to re-price the product offered by the user to a price higher or lower than a price of a product in the first analysis result.

16. A non-transitory computer-readable storage medium having computer-executable instructions stored thereon which, when executed, cause operations to be performed, in which the operations include:

receiving, via a user computer, over a network, a first user selection including a product, and a brand, vendor and category associated with the product;

accessing a set of product information that includes information regarding the selected product, the set of product information including a set of products, wherein the product information includes for each product in the set: one or more brands under which the product is sold, a set of vendors offering it for sale, and a product category;

generating a set of analysis results based on the obtained product information;

identifying a first statistical pattern in a first analysis result from the set of analysis results for the selected product by performing statistical analysis on the product information and the analysis results, in which the first statistical pattern is a statistical pattern between two or more types of data included in the first analysis result or in the product information and in which the two or more types of data share a common aspect, and in which the first statistical pattern comprises a cyclic change in the first analysis result,

in which the set of analysis results includes:

a social metric, an identification of which product in a sub-set of products leads or follows other products in the sub-set of products in terms of price changes, a demand metric based at least in part on visitors record generated from webpage traffic to one or more online stores at which the product is available, and in which the demand metric is stored in the database coupled to the computer, and a reach of the product in terms of the number of people who visit an online sales venue of the product, in which the social metric is generated based on a number of followers or a number of likes of the product, and in which the social metric is stored in a database coupled to the computer;

transmitting to the user computer, over the network, for display to the user, the first analysis result and the first statistical pattern;

receiving, via the user computer, over the network, a create alert command for the product, the create alert command including:

alert criteria, including at least one of: absolute or percentage change in price, initiation or termination of sales at a venue, and

a notification window and frequency;

in response to the command, executing the alert and transmit to the user computer, over the network, a notification;

receiving a second user selection and a second analysis result with respect to the second user selection; and

identifying a second statistical pattern in the second analysis result.

17. The non-transitory computer-readable storage medium of claim **16**, in which the first user selection further comprises at least one of a user list and a user favorite.

18. The non-transitory computer-readable storage medium of claim **17**, in which the user list comprises a custom list or a smart list.

19. The non-transitory computer-readable storage medium of claim **18**, in which the smart list is created upon receipt of a selection of at least one of a brand, a vendor or a category in the set of product information and at least one criterion for limiting the first analysis result presented in relation to the user list, and in which the custom list comprises a list of products provided by the user.

20. The non-transitory computer-readable storage medium of claim **16**, in which the operations further include:

comparing the second analysis result relating to the second user selection to the first analysis result, and

making a first recommendation to the user based on the comparison of the first and second analysis results.

21. The non-transitory computer-readable storage medium of claim **20**, in which:

the first analysis result comprises, for a product sold by a first vendor: a demand metric for the product sold by the first vendor, a price history for the product sold by the first vendor, and a promotion metric for the product sold by the first vendor;

the second analysis result comprises, for a product sold by a second vendor, a demand metric for the product sold by the second vendor, a price history for the product sold by the second vendor, and a promotion metric for the product sold by the first vendor; and

in which the first recommendation comprises at least one of:

a recommendation to add or remove the product sold by the first vendor or the product sold by the second vendor from an inventory of the first or second vendor,

a recommendation to the first or second vendor to charge a higher or lower price for the product sold by the first vendor or the product sold by the second vendor, and a recommendation to the first or second vendor to increase or decrease a promotion of the product sold by the first vendor or the product sold by the second vendor.

22. The non-transitory computer-readable storage medium of claim **21** further including:

determining if the demand metrics in the first and second analysis results are low;

and in response to a determination that they are low, recommending to the user that the product sold by the first vendor or the product sold by the second vendor be removed from inventory.

23. The non-transitory computer-readable storage medium of claim **21**, further including:

when:

the price histories of the product sold by the first and second vendors indicate a price difference that exceeds a pre-defined threshold, and

there is a difference in the promotion metrics between the product sold by the first and second vendors, at least one of:

recommending an increase in the price of the product sold by the first vendor or the product sold by the second vendor to a vendor associated with a higher demand metric, a lower promotion metric and a lower current price,

recommending an increase in the price of the product sold by the first vendor or the product sold by the second vendor or a lower promotion to a vendor associated with the higher demand metric, a higher promotion metric, and the lower current price, and

recommending that the product sold by the first vendor or the product sold by the second vendor be discontinued by a vendor associated with a lower demand metric, the lower promotion metric, and the lower current price.

24. The non-transitory computer-readable storage medium of claim **16**, further including:

receiving from the user, over the network, an instruction to link two or more different products in the set of product information; and

in response to a number of times the two or more different products in the set of product information are linked by any user exceeding a threshold, identifying the two or more products as substitutes for each other.

25. The non-transitory computer-readable storage medium of claim **16**, further including:

receiving an instruction to send a notice to the user when the first analysis result corresponds to a user-specified alarm setting.

26. The non-transitory computer-readable storage medium of claim **25**, in which the user-specified alarm setting comprises at least a first target in the first user selection and one or more alarm criteria.

27. The non-transitory computer-readable storage medium of claim **26**, in which the one or more alarm criteria comprise at least one of:

an absolute or percentage change in price of the first target,

a promotion of the first target, and

an inception of availability or discontinuation of availability of the first target.

28. The non-transitory computer-readable storage medium of claim **26**,

in which the one or more alarm criteria comprise a second target comprising at least one of: a product, a category in a product categorization schema, a brand, and a vendor, and

in which the one or more alarm criteria comprise at least one of an absolute or percentage change in price between the first and second targets and a change in the promotion status of one of the first and second targets relative to either the first or the second target.

29. The non-transitory computer-readable storage medium of claim **16**, in which the operations further include:

associating the user with a competitor of the user selection and in which presenting the analysis result comprises presenting the first analysis result in relation to the user selection and the competitor.

30. The non-transitory computer-readable storage medium of claim **16**, in which the operations further include:

receiving a re-price command from the user, and, in response, reprice a product offered by the user in response to a condition of the first analysis result, and

in which the re-price command comprises at least one of an instruction to re-price the product offered by the user to a price higher or lower than a price of a product in the first analysis result.

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